

UNITED NATIONS ECONOMIC COMMISSION FOR EUROPE

LOCATION EFFICIENCY AND ACCESSIBILITY OF E-MOBILITY

Mobility needs and infrastructure planning



UNECE

UNITED NATIONS ECONOMIC COMMISSION FOR EUROPE
Group of Experts on Energy Efficiency

Location efficiency and accessibility of E-Mobility: mobility needs and infrastructure planning

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In 2024, in recognition of the need to align the deployment of electric vehicles with the development of related charging infrastructure and the electric grid, taking location efficiency, systemic efficiency, and energy system resilience into consideration, the United Nations Economic Commission for Europe (UNECE) established an Informal Task Force on Electric Mobility – a multidisciplinary and cross-dimensional initiative.

Recognizing the complexity of energy-transport interlinkages in the context of sustainable mobility, the mandate of the Informal Task Force on Electric Mobility for 2024–2026 outlined in ECE/TRANS/2024/21, comprised a set of thematic clusters. The discussions during the first session of the Informal Task Force on Electric Mobility in 2024 and the subsequent online questionnaire, led to the selection of four priority clusters/subclusters for inclusion in the work plan for 2024–2025, including Cluster E titled “Explore the concept of location efficiency and accessibility in terms of infrastructure planning and reducing mobility needs by linking buildings, industry, transport, and infrastructure through land use” overseen by the Sustainable Energy subprogramme and, in particular, the Group of Experts on Energy Efficiency. This document presents the research outcomes of the Cluster E work in 2024–2025.

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¹ Informal Task Force on E-Mobility, Cluster E, Expert Consultations “Location efficiency and accessibility: infrastructure and mobility needs: (online, 06 March 2025). See: <https://unece.org/info/Sustainable-Energy/Energy-Efficiency/events/399887>

² See: <https://unece.org/sustainable-energy/events/unece-sustainable-energy-week-2025>

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LIST OF ACRONYMS AND ABBREVIATIONS

AC	Alternating current
AI	Artificial Intelligence
BEVs	Battery electric vehicles
CPOs	Charge points operators
DC	Direct current
DUT	Driving urban transition
EMIC	E-Mobility Information Centre
EU	European Union
EV	Electric vehicle
ICE	Internal combustion engine
MaaS	Mobility-as-a-service
MM	Micro-mobility
NEVI	National electric vehicle infrastructure
PT	Public transport
PV	Photovoltaics
TCO	Total cost of ownership
V2G	Vehicle-to-grid
VKT	Vehicle kilometres travelled

EXECUTIVE SUMMARY

Policymakers and stakeholders will find in this report key insights, structured interpretations, case studies, and actionable suggestions, building upon the experience in UNECE countries and beyond, for systemically efficient developments of electric mobility.

Electric vehicles (EVs) are being increasingly adopted globally, and charging infrastructure is expanding alongside with it. However, EV sales and adoption still need to grow further to substantially contribute to current climate-goals. EV adoption does not only require sufficient renewable energy, EVs, and the right number and types of chargers, but also that EV charging infrastructure is optimally located where it is needed. It also requires a recognition of the importance of collaborative governance and effective communication to (future) consumers to enhance the effectiveness of information sharing, public participation, and legitimacy of the EV transition. To ensure that these interconnections are considered when thinking about EV adoption, this study proposes an innovative framework for understanding sustainable mobility by highlighting the interconnectedness between e-mobility, location efficiency, and governance.

This report explores multiple locations for charging infrastructure and discusses various conditions under which they improve or hamper systemic efficiency. Moreover, it examines which policies can effectively contribute to EV adoption, how the expansion of charging and grid infrastructure can be financed smartly, and where charging infrastructure is most effective in promoting EV adoption. The report explores multiple examples of projects and cities that enabled strong public participation in the context of e-mobility. It also addresses communication and outreach strategies and tactics to effectively promote the EV-adoption.

A crucial part of the EV transition is to ensure that EVs and associated technologies are accessible to and adopted by (future) consumers. For this reason, the report outlines a wide range of policy measures that can stimulate certain behaviours and facilitate the adoption of system-efficient e-mobility. The report also introduces diffusion theory, as well as the different phases associated with a market's penetration of EVs. Recognizing these different phases is critical to devise appropriate policy measures for each of the phases.

Attention is paid to the wide variety of national circumstances that portray the UNECE region, with the view that one-size-does-not-fit-all and that country-led programmes need to build upon careful considerations of such (and further) circumstances, in a way that would have, however, been impossible to systematically explore. Each country needs to find its own way to cope with the challenges, aware that other countries may have been pioneering solutions or meeting similar obstacles.

Thus, in short, this report explores how e-mobility can be (i) placed efficiently to improve accessibility, (ii) communicated effectively, (iii) implemented with justice, and (iv) contribute to net-zero climate-goals, as essential components of a just transition for sustainable and resilient energy.

INTRODUCTION

E-mobility, location efficiency, and governance: thinking them together

Systemic efficiency of e-mobility can frame and address the challenges (and maximize the opportunities) related to a technological trajectory that has a large potential for making mobility more sustainable but also may require steering.

The spontaneous trajectories may produce vast inequalities across sub-national regions and even across countries, with temporal gaps in diffusion, uncertainties and stop-and-go dynamics.

Having a clear idea of where to position the different territorial requirements of this new mobility, fitting a multiplicity of requirements deriving from the grid, the operators, the users, the spatial planners, and a multiplicity of other stakeholders would help stabilize expectations, minimize the investment costs (and usage of space), speed up adoption, integrate a plurality of vehicles (beyond e-cars).

Systemically efficient e-mobility responds not only to technological constraints but also fits an evolving social system, with inclusion and resilience at the core. Accordingly, in this report policies appear in several chapters in connection to stakeholders' strategies and actions.

In a concise graphical manner, the systemic efficiency and accessibility of e-mobility can be seen as the result of the convergence of six key factors, as depicted in Figure 1.

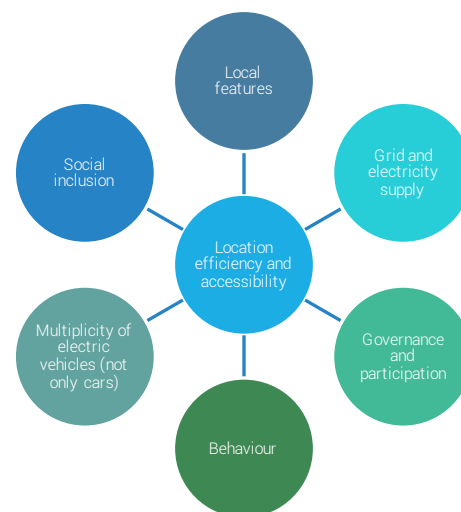
A system of locations is systemically efficient and accessible if it is integrated with other uses (including current parking and road access), is adequately connected with the grid (supplied with zero-carbon electricity), can be used by a wide variety of vehicles (including low cost options, such as e-bikes), is socially inclusive (in terms of prices and proximity also to low-income neighbourhoods), fits the behaviours of users (who are dynamically learning about e-mobility), and has required not too much investment by Charging Points Operators. All this requires stakeholder sensitivity, engagement, and actions, supported or (in certain cases) orchestrated by national and subnational public administrations.

Conversely, a system of locations is not systemically efficient and accessible if all the chargers are concentrated in a sub-region, leaving large areas without opportunities, the utilization is low and limited by incompatible standards (including for payments), with long queues during vacations and other peak moments, the connection with the grid is insufficient, and power is generated with high carbon intensity.

Detailing this vision, this structured and comprehensive text is intended to explore options and offer venues of reflections for policymakers and stakeholders in their strategic thinking but also in some of their practical tasks.

This study proposes an innovative framework for understanding sustainable mobility by highlighting the interconnectedness between e-mobility, location efficiency, and governance with

Figure 1
Concept of location efficiency and accessibility



the view of attaining systemic efficiency. Enhancing the overall efficiency and accessibility of an e-mobility system require location-efficient solutions. This means that successful EV adoption does not only require sufficient renewable energy, EVs, and the right number and types of chargers, but also about ensuring that these elements are optimally located where they are actually needed.

Moreover, the overall efficiency for EV transportation is influenced by multiple factors, such as infrastructure availability, charging stations, road networks, renewable energy integration, urban density, norms and values, social practices, local policies, etc. Depending on the specific location, the relative impact of these factors on the overall efficiency varies, meaning the optimal approach to achieving the highest location efficiency in EV transportation differs from place to place. For example, even if EV charging infrastructure is located in a place that, in principle, reduces travel distances for the region, it might not be widely known or may not fit well into regional social practices and routines.

Location efficiency refers to the strategic placement of critical and daily infrastructure and amenities to increase accessibility and energy security by reducing travel distances. In the context of e-mobility, location efficiency refers to the strategic placement of charging infrastructure. The availability and strategic placement of charging infrastructure are key to making e-mobility accessible to potentially everyone and preventing transport poverty – the persistent lack of accessible mobility options – as well as social and economic exclusion. Given that places differ in the extent to which they strategically locate charging infrastructure to increase accessibility and energy security, it is worthwhile comparing common locations (see later discussion). The challenge, however, is not only to increase the sales in EVs and ensure location efficient charging infrastructure, but also to enable an effective integration into the power grid and technologies that support grid resilience (vehicle-to-grid, V2G).

The third component of this framework relates to the less technical aspects of the EV-transition, that focuses on governance, policy coordination, effective communication and outreach strategies, regulations, incentives, and public participation. Governance does not only need to consider performance and ways to decrease bureaucratic hurdles and accelerate policy and technological deployment, but also needs to be oriented towards collaboration, deliberation, and including relevant and diverse stakeholders. A more communicative and collaborative mode of governance can increase legitimacy, acceptance, satisfaction amongst participants, and potentially the effectiveness of devising collaboratively arranged location efficient EV solutions. Thus, in short, this report explores how e-mobility can be placed efficiently to improve accessibility, how it can be communicated effectively, implemented justly, and contribute to net-zero goals.

Discussions on these topics often exhibit a dichotomy between specific case studies with a regional focus and abstract guidelines that must be transposed into specific regions. We consider it essential to take both perspectives into account. We do so by exploring multiple typologies of situations and case studies, across different geographies, including different stakeholders, modes of EV transportation, and factors promoting location-efficient, inclusive e-mobility changes.

Nonetheless, the political, social, technological, and environmental differences across geographies make it impossible to provide universal and simultaneously hands-on guidelines. The goal, thus, is to consider a multitude of relevant factors that are conducive to more location-efficient, inclusive e-mobility solutions that are relevant across geographies. Also, we consider multiple modes of EV transportation that go beyond the most prominent EV modes and stakeholders. Taking on such a broad perspective on these topics is critical to devise effective strategies to achieve net-zero targets.

Bridging location efficiency and e-mobility

Location efficiency and electric mobility (e-mobility) convergences presents a unique opportunity to redefine the future of sustainable urban development. Location efficiency, which refers to the optimal spatial organization of housing, employment, and essential services to reduce the need for long-distance travel, directly complements the goals of e-mobility by enabling cleaner, shorter, and more energy-efficient transport choices. Together, these two concepts form partly the backbone of low-emission, accessible and people-centered urban mobility systems.

As cities grapple with congestion, air pollution, and climate vulnerability, the deployment of EVs need to be thoughtfully integrated into land use and transport planning. Poorly coordinated charging infrastructure, for example, can exacerbate spatial inequality and reinforce car dependency. Conversely, when EV infrastructure is co-located with mixed-use developments, transit hubs, and non-motorized transport networks, it enhances connectivity, encourages behavioural shifts, and reduces overall vehicle-kilometres travelled (VKT).

Location-efficient e-mobility strategies not only optimize energy consumption but also unlock cross-sectoral benefits. Smart charging systems, when situated in high-accessibility zones, improve grid performance and support renewable integration. Similarly, aligning EV deployment with compact development patterns reduces pressure on road networks and parking demand, freeing up urban space for green infrastructure, public amenities, and walkable environments.

However, the alignment of location efficiency and e-mobility requires deliberate governance, institutional coordination, and policy coherence. Urban zoning regulations need to account for future-proof mobility needs, building codes need to require EV-ready design, and public-private partnerships require prioritize infrastructure equity across all neighbourhoods.

Besides, bridging location efficiency and e-mobility demands a cross-sectoral approach. Urban planners, energy providers, transportation agencies, and private stakeholders need to collaborate to design systems that reflect real-world user behaviour, land use configurations, and future growth trajectories. Key considerations include identifying optimal charging station locations based on traffic patterns and residential densities, integrating renewable energy into charging networks, and leveraging digital platforms for real-time monitoring and user navigation.

Furthermore, achieving a just transition requires addressing disparities in e-mobility access. Many underserved or peri-urban communities may lack both charging infrastructure and the locational advantages that support low-carbon transportation. Policies that incentivize EV infrastructure in high-density, transit-served locations, as well as in low-income and marginalized areas, are vital to closing this gap.

Generally, the interface between location efficiency and e-mobility offers a transformative opportunity to reshape urban mobility systems. By leveraging spatial planning tools and fostering integrated infrastructure development, cities can support a cleaner, smarter, and more resilient transport future. This chapter explores this nexus, highlighting strategies, case examples, and policy frameworks that can bridge the two domains effectively.

Governing mobility: participation in the context of location efficiency and e-mobility

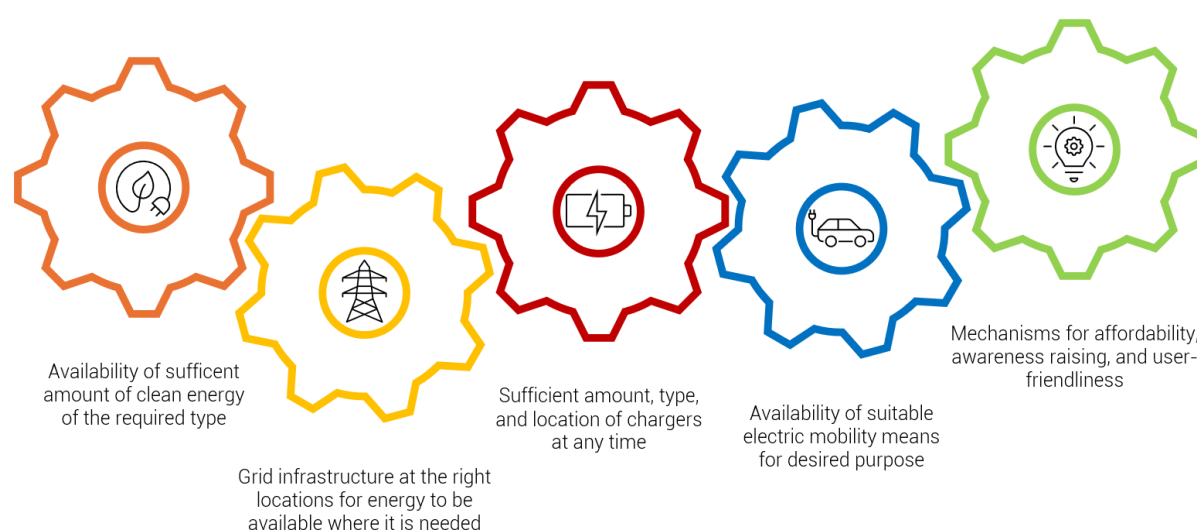
Considering the governance of sustainability transitions for location efficient EV-transportation is critical to ensure both an effective and just transition, which is key for the success of the transition itself (UNECE, 2025). We refer to governance as the coordinated and integrated interactions and actions between diverse stakeholders to address societal challenges (Kooiman, 2003). This definition illustrates our focus on the process of political action and public participation as well as the integration of policies pertaining e-mobility systems and location efficient solution. Policy

integration ensures consistency and effectiveness by aligning policy goals, instruments, subsystems, and frames (Candel and Biesbroek, 2016). This approach constitutes the opposite of silo-thinking and fragmented developments in location efficient EV planning. Policy integration necessitates strong communication between stakeholders and a strong involvement of them. Although the types and number of direct EV stakeholders is relatively small, the public participation can address lack of interest or even resistances to e-mobility transition measures.

While improving the communication between diverse stakeholders in e-mobility systems seems both critical and perhaps obvious, what is less apparent, however, is which methods and formats are both most appropriate in ensuring location efficient e-mobility interventions. While some venues and proposals are contained in the dedicated Governance section, successful implementation of e-mobility necessitates taking the following five key areas into account simultaneously (Figure 2).

Figure 2

Key areas for implementation of e-mobility



This report explores how e-mobility can be placed efficiently to improve accessibility, how it can be communicated effectively, implemented justly, and contribute to net-zero goals.

KEY CONCEPTS AND CONTEXTS IN SUSTAINABLE MOBILITY

Beyond just cars: exploring e-mobility transportation modes

This chapter concisely contains the following elements:

1. List of types of EVs
2. List of parameters that describe the EVs (e.g. range, battery size)
3. Comparison across different vehicles
4. Car-centricity vs multi-modal organization of mobility
5. Hints about systems of e-mobility transportation modes (vehicles, infrastructure, operators, regional and super-regional coordination)
6. Electrification as paradigm embracing (but also going beyond) mobility

With many countries and regions across the world aiming towards the goal of climate neutrality, considering multiple modes of electric mobility becomes critical. The more so if the vehicle is appropriate to the mobility needs and has several co-benefits (such as noise abatement, air quality), with their cascading health benefits.

Across this paper, a broad and inclusive definition of e-mobility is adopted, i.e., all types of current and future means that are not immobile and require electricity to transport people or goods or execute a service while changing location. Traditional electric mobility includes:

- Metro/underground systems
- Tramways
- Elevators/escalators/walkways (tapis roulant)
- Trains (since 1837)

In modern times, electric mobility further encompasses:

- Electric micro-mobility (MM), including hoverboards, electric skateboards, and other small personal vehicles
- E-bikes (including pedelecs and throttle-controlled e-bikes)
- E-scooters
- Intermediate light e-vehicles (three- and four-wheelers)
- E-cars
- E-buses and e-coaches
- E-trucks (for commercial, city-logistics, and long-distance purposes)
- Electrified trailers and further uses of in-wheel motors
- Vocational vehicles (electrified loaders, excavators, dumpsters, construction and municipal service vehicles, mowers, etc.)
- Other modes of electric mobility, such as maritime (e-boats, e-ships, etc.) and airborne (drones, electric vertical take-off and landing aircraft (eVTOLs), unmanned aerial vehicles (UAVs), etc.) that are beyond the scope of this study.

All these vehicles require specialized or common systems for routing and charging (both with and without batteries), offering a highly flexible and competitive alternative to fossil fuel-based transportation. Some of those systems have a long history, others are still in their infancy for many countries. In the future, electricity might be powering the movement of objects whose shape drastically innovates on today's designs. For example, in India, where three-wheeled rickshaws are prominent, registrations in electric three-wheelers soared from five thousand to 580 thousand within 8 years (2015-2023) (IEA, 2024).

These vehicles are not only purchased by households for private trips, but they are purchased by companies and logistics operators in fleets, for multiple purposes, including public and shared transportation. In this vein, it is particularly relevant the paradigm shift of vehicles, digital infrastructure and legislation leveraging ACES: Autonomous, Connected, Electric, and Shared mobility, with its perspective of safety, simplification, emission reduction.

Electrification is a technological paradigm extending in many directions beyond mobility (e.g. computers, telecommunication, home automation, etc.). Some electric tools perform both work and movement in domestic settings (such as electric and autonomous lawn mowers, robot vacuum cleaner, etc.) and industry-specific settings (such as excavators, loaders, etc.).

The typical parameters by which to compare EVs are the battery size, the range, the time to charge (0%-100% or 20%-80%), and the added range by 1-hour charge. A key intermediate indicator is the aerodynamic efficiency and the weight, leading to overall efficiency in energy consumption.

In addition to these technical parameters, often in the hands of manufacturers, it is important to broaden the comparison criteria to get to systemic efficiency. From a societal perspective, vehicles that are lightweight, connected and with additional innovative safety features are particularly relevant and preferable, e.g. to reduce accidents and their impacts. Smaller vehicles occupy less space, allowing for more throughput and less traffic. They may be more compatible with walkability and pedestrian areas, characterized by vibrant societal and commercial life. In particular, if the vehicles can be transported within trains, buses and other PT means, they foster multimodality, which tends to be systemically more efficient than the car-centric mono-modality. Whether the vehicles can be easily stolen (or not) is also relevant, since, for instance, e-bike at the high price end range have very good performances, becoming a good substitute for many car-based trips, but some people are afraid of the risk of theft (and lack of affordable insurance). Accordingly, policies and technologies that reduce theft and other safety risks will contribute to the even larger take-off of e-bikes and other e-MM tools, characterized by very high energy efficiency and low space occupation.³ To assess the overall efficiency and effectiveness of EVs, it is critical to include these terms in the analysis.

Location efficiency and its various contributing factors, including the integration with buildings and industries

Spatial concentration of origins and destinations of mobility flows can reduce the need for long and frequent trips. *Mobility is needed only when the activity is not co-located and cannot be accomplished through telecommunication*: by reshaping activities and their location, it is in principle possible to reduce person-km. In the Middle Ages, this was obtained with feuds (when all activities were strictly local) and with compact cities surrounded by walls. Long-range (horse-powered) trade was involving only a thin social group of merchants. With the Industrial Age, the infrastructure availability and the fall in transport cost, thanks to steam, petrol and diesel engines on land and water, as well as non-electric and electric trains and finally aviation, contributed to the globalization of freight and personal mobility, with an explosion of VKT. The unhealthy conditions around factories prompted “single-use zoning”, with planned and actual separation of residential, industrial and service areas (with the corresponding need of mobility across them). The densification of the towns and industrial districts and the predictability of workers' flow at common hours provided the background for public transport (PT).

³ For higher safety of micro-mobility and vulnerable road users, see: <https://phoebe-project.eu/> and <https://www.events-project.eu/road-safety-cluster/>

With the Third (if not Fourth) Industrial revolution, light industry and the dominant office environment are more compatible with residence, thus prompting a movement towards mixed-use of land, with the hope of reducing mobility needs. However, the fragmentation of employment into Small and Medium Enterprises, temporal asynchronous working times, with the rise of part-time and intermittent work, as well as less stable work (with the end for many of a “lifelong workplace”) and family relationships in which both spouses work (leading to need of spatial compromise between two often different location of work), with rising divorce rates, have largely put in crisis the traditional location efficiency (and traditional planning of PT).

With the key importance of location efficiency and public transport effectiveness in reducing emissions and increasing livable human settlements, new approaches are currently experimented:

- strategic urban planning with actions inducing mixed-use zoning with residency, light industry, offices, commercial areas, education, amenities;
- the 15-minutes-city, decentralizing and supplying several categories of everyday services in a range of 15 minutes walk;
- urban transformation emphasizing walkability, complete streets for users of all types, abilities and vulnerabilities, including those suggested and tested in the Driving Urban Transition (DUT) programme embracing more than 20 countries in the UNECE region;
- timetable and optimizing Apps fostering use of multimodal and public transport;
- geospatial modelling, in which competitiveness indices and spatial regression identify underserved areas, guiding facility placement and policy interventions to balance accessibility;
- further green digital solutions, both at user level and system level (such as digital twins and open-source models like SUMO – Simulation of Urban Mobility), often the result of cooperation of private actors and collective intelligence through Living Labs;
- on-demand and flexible public transport routes;
- a resurgence of tramway and other electric collective means of transport;
- Mobility-as-a-Service (MaaS) platforms bundling various transport modes, often incorporating dynamic pricing and incentives to optimize modal efficiency and urban land use.

In this context, this report of the Informal Task Force on E-Mobility is particularly interested in the contribution to location efficiency and accessibility of e-mobility.

In some geographies, these concepts are already lived experiences. For example, in Germany, almost 2000 municipalities enable their citizens to reach critical places within 15 minutes (Die Zeit, 2025). However, many other places are still in the early stages of implementing location efficiency concepts, particularly location efficient e-mobility solutions.

To address this, in this chapter, we cover the localization of EV recharging stations, while subsequent chapters and sub-chapters address the types of charging systems, the behaviour of users (thus, the localization of vehicles over time), and how to mobilize investments in the most efficient locations.

The localization of EV recharging stations (be they private or public) is essential to the location efficiency of e-mobility. A location may be more or less well connected to the grid or to an off-grid energy supply; it can be easily accessible for the vehicles or not; it may require long detours from optimal pathways; it may be visible (thus impacting the perceived presence of recharging opportunities) or not; it can be congested or not; it may prompt additional activities during the charge; it may be frequented mostly in certain hours of the day and the week, impacting on when the recharge takes place, thus, in case of certain electricity mixes, the type of electricity supply used

(and the corresponding emissions and wholesale prices). The location may suggest the typical speed of the charge.

Other factors that influence the location efficiency is the alignment with social practices, habits, and routines (i.e., shopping practices, parking behaviour) of EV users. Thus, location efficiency is enhanced if it is properly aligned with daily consumer behaviour. In addition to that, location efficiency can be enhanced by integrating charging stations into existing infrastructure (e.g., lamppost charging, underground garages, parking slots). Another factor is the demand for EV charging stations. Thus, a location is suitable for EV-charging stations if there is a great enough demand from EV users. Finally, location efficiency is greatly influenced by the degree it promotes accessibility. Physical access, user conveniences, and grid and power accessibility have major impacts on location efficiency.

To broadly explore the many possible locations and allow subsequent judgement of priority and alignment with different country structures, phase of the EV diffusion process, types of vehicles recharged, we shall present in a table a relatively long list of locations, with the conditions under which they are efficient, the reasons for which they may not be efficient, which are the obstacles to deployment of EV charging points in such locations and what can be done (by policies and stakeholder actions) to deploy there, fostering the fulfilment of the conditions of efficiency.

Table 1 assumes that charging takes place at the location in the same moment as the grid is loaded, thus the typical time in which vehicles can be expected to be at the location reverberates on the electricity mix of the charge and its carbon intensity.

However, it should be noted that in most, if not all, locations a battery could be added so to decouple the two events both in time and in quantities. This increases significantly the resilience of the system to outages, may reduce the peak load required, thus the cost of the electricity, as well as the induced emissions. As the cost of batteries continues to fall (including with chemistries with low or even without rare earth elements and lithium) and a new flow of second life batteries might be planned and managed, this solution of battery-buffered EV charging can ramp up.

Table 1 is mainly meant for conventional grid-connected recharging pools, stations and points. It can be applied also to battery swapping stations which have the additional efficiency of low charging time, of decoupling of the moment in which the grid is loaded from the moment in which the battery is installed in the vehicle, and of fostering the adoption of a flexible mix of battery dimensions (small for everyday uses, larger for longer trips), which in turn reduces the upfront cost of the purchase and is systemically efficient (smaller batteries mean faster grid-connected charges, lower quantities of rare and critical minerals, etc.). Battery swap stations have typical locations as presented in Table 2.

Table 1 – Location of EV charging points and their efficiency conditions and disturbances, obstacles to deployment and what can be done

<i>Location of EV charging point</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Home (independent house)	Highly compatible with user timetable and the grid capacity. The cheapest and most comfortable way from the point of the user. It pools charging with normal consumption by the household, with no requirement of grid update. It favours a rooftop photovoltaic (PV) installation, for which it cuts the payback period in less than half.	Charging time tends to be in the evening, in case of a daily recharging pattern. Accordingly, if the house does not have a PV + battery system, the charge adds to the evening peak load of the grid, with, at high penetration stage of e-diffusion and depending on national and local conditions, some negative impact on grid congestion and emissions. Normal wall socket enough for basic charging but wallboxes necessary for more regular charging.	Costs and permitting. In some countries, a permit is not needed up to 11 kW, but one should tell the grid operator about installation. A permit would have to be obtained before installing a wallbox with 22 kW or higher power.	Subsidize the wallbox, especially if publicly accessible (to peers in a club logics or to the general public) for free or a fee. Timers, scheduling, remote monitoring, energy management, bidirectional charge V2H, V2B can be promoted.
Home (multistorey building, both in ownership and rented)	Highly compatible with user timetable.	Electricity usually taken from condominium-level grid, with lack of attribution to the actual charger. In the high penetration stage, several high-speed charging may be a significant building-level load for the local grid.	Disagreement with co-owners. Parking space may not be available or attributed. In case the flat is rented, then the landlord may oppose or refuse to pay for the wall outlet (and the tenant may be wary of making investment without certainty of a multi-year use). Uncertainties as for whom pays for the charge.	Establish a right-to-charge, independent of co-owners and landlord will. Mandatory rules to allow charging if building offers parking space. Smart charging distributing the charge across all charging points (ideally available at all parking slots). Include the charging costs in the rental. Local and regional authorities should encourage co-owners to conduct fire risk assessments of their buildings equipped with recharging infrastructure to ensure a high level of safety for EV users and minimize the risk of fire.

<i>Location of EV charging point</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Paid parking space in residential area	Compatible with parking habits.	Charging may be expensive and reduce the total cost of ownership (TCO) advantage of EVs. Penalties on vehicles having terminated charging may be levied during the night, when removal of vehicle is impractical.	Parking operator may decide not to provide charging (out of inertia, disinterest or actual opposition).	Mandatory standard for paid parking spaces above a certain number of slots to provide charging, at well above the current share of EVs in the vehicle fleet. Include the charging cost into the payment for parking. If mandates cannot be adopted, then wide campaign of recruitment of willing parking operators and of communication about charging opportunities in their premises. If dynamic panels inform about available slots, then insist on evidencing charging slots.
Free parking space in residential area	Compatible with cheap parking habits. Possible synergy with public lightning system.	Random allocation of space.	Lack of interest in providing charging by the owner of the road and of the parking area (often not clearly identifiable by the user).	Charging installation by request to municipal authority for free or partially paid by the requesting drivers.
Hotel and other tourist / business accommodations	Highly compatible with customers' timetable. Parking facility already existing. Possible pooling of payment with stay price and system.	Accessible only to customers. Possible seasonality in utilization.	Low awareness of guests. Hotel owners may delegate to nearby charging point, without assuring quality and assistance to their clients.	Including charging in the definition of hotel classification (e.g. 3-stars). Leverage the analogy with Wi-Fi, which was first considered a premium feature in good hotels and now is standard in any type of accommodation. Involvement of tourist intermediaries.
Multifunctional (mixed-use) area	Multiple users at different times of the day. High occupancy rate. Integration of charging with current activities (work, leisure, residency) for all ages and genders. Easy to find activities to carry out during charging. Supportive of the vibrant walkable area.	Traffic to charging point can congest the area or disturb its walkability.	Lack of keen stakeholder interested in locating the charger. Opposition by current users of the area.	Carefully crafted location at the micro-level for access and compatibility. High aesthetic premium. Local campaign to elicit current and solicit future EV use.

<i>Location of EV charging point</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
City centres	High EV user demand, support for air quality goals, multiple opportunities for integration with existing infrastructure (e.g., streetlights, underground garages)	Spatial confinements for additional charging infrastructure; comparatively expensive installation due to land costs	Difficulty in allocating public space in dense urban areas for chargers	Integrate charging station into existing infrastructure; implement pricing strategies to ensure turnover of charging spots. Utilize non-conventional but highly compatible shapes for charging (e.g. integrated at lampposts). New recharging technologies such as battery swapping could be deployed to enable EV users to charge quicker whereas swapped batteries can be recharged at the stations during off-peak hours to avoid grid congestion (see Table 2).
Charging point at the service of pedestrian areas	Supportive of a broader zero-emission zone surrounding the pedestrian area. Supportive of active mobility, e-sharing, sustainable values.	Risk of low occupation rate. Useless if the pedestrian area will be extended to include the charging point.	Opposition by current users of the area. Silos mentality in municipality department.	Embedding e-mobility in sustainable city discussion. Inclusion of e-bike charging.
Intermodal hub (train station, metro station, airport, port, etc.), in the logics of park-and-ride (P+R)	Prompting PT use, providing an incentive to stop at the intermodal hub instead of reaching the final destination by car. If for free, cost incentive to PT users.	Little relationship between duration of charging and of returning trip (from PT). Usually charging is shorter, which means that the vehicle is connected while not in charge. If this is penalized with a fee, then very punitive for the user.	Low profitability for the Charge Point Operators (CPOs). Permitting from the PT.	Smart charging with a large number of slow charging points (typically AC chargers), without penalization for terminated charge. Valet service (with car keys dropped to a person that will charge the vehicle and then disconnect and move it out of the charging slot) (job creation, higher cost).
Secondary road	Low congestion, easy access. Multiple moments of the day in which it can be used.	Lack of awareness of its existence if not from digital tools. Low utilization.	Low priority given to the place, possibly linked to low profitability. More radically, the location may simply not be considered.	Parking area to be signaled.

<i>Location of EV charging point</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Parking spaces connected to primary roads	Compatible with regular road trip habits and high EV user demand. Normally well connected to the grid as they are not far away from main electric lines.	As a result of high congestion, longer waiting times for recharging. Due to spatial limitations in the parking spaces accessing a primary road, chargers may not easily meet the minimum accessibility requirements in terms of adequate free moving space around the charger, and barrier free access for the people with reduced mobility. Victim of vandalism of the recharging infrastructure, namely cutting cables or breaking screens.	Value or opportunities in such deployment is oftentimes not seen.	Governments should set non-binding minimum dimension requirements for charging infrastructure and sites to improve user accessibility and foster market competition between operators that comply with these requirements and those that do not. Maximization of space efficiency at the charging stations through forethought planning of the location of streetlights, waste bins, and restrooms is a good practice for CPOs. Install more stalls per charging station to manage congestion and ensure the continuation of charging service at the time of a down stall. To mitigate the risk of vandalism and ensure charging stations remain reliable, one of the preventive measures is to place stalls in well-lit, high-visibility areas under the surveillance of high-quality CCTV cameras with real-time monitoring. Another means of preventing vandalism is to raise awareness in the local communities to report suspicious activities.

<i>Location of EV charging point</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Highway rest areas	Fundamental for long trips without inefficient detours. Equally important for e-cars, e-coaches and e-trucks. Highly visible, so taken into consideration by wary potential buyers of EVs (who would notice the lack of them or the long lines in front of them is they are insufficient and congested).	High power recharging pools may or not be well connected to the grid. They may be far away from main electric lines (in certain countries). As installation may precede e-mobility take-off in sales, large investments may have a long payback period. If the operator decides to have very high prices to quicken payback, then there is an erosion of the TCO advantage of EVs.	Monopoly on each area and oligopoly on the networks of areas, highly interested in sales of fossil fuels, can prevent installation (or impose certain operators).	Mandatory standards prompting installation. Transparent tendering of the operators. The hardware of charging stations should be designed to resist severe weather, extreme temperatures, and potential human damage to the charging equipment. To ensure reliability, EV chargers should undergo regular maintenance and inspections to check for damage, cleanliness, and any obstructions like snow or trash. To enhance accessibility, governments should specify minimal site requirements for user accessibility to be complied by the CPO or location owner at the charging pools with non-exclusive priority charging for people with reduced mobility while CPOs are given 'freedom of site design' to optimize user experience. Non-exclusive priority charging is placeholder-based access to charging instead of exclusive charging or restrictive access to one single user group, with priority for all persons with reduced mobility. Governments should grant priority grid access to net-zero technologies like EV charging infrastructure in order to avoid disruptions or delays with recharging pools getting connected to the power grid. Governments should streamline and digitalize permitting procedures for grid connection and require from system operators to provide transparent grid capacity data to the investors. To response grid congestion, anticipatory investments is an important step for enabling grid renovations of offshore networks and untapped onshore grids. New recharging technologies such as battery swapping could be deployed to enable EV users to charge quicker whereas the swapped batteries can be recharged at the stations during the off-peak hours to avoid grid congestion (see Table 2).

<i>Location of EV charging point</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Gas stations wherever located	They maintain the refuelling habits of ICE drivers and reuse their facilities and surrounding amenities (e.g. food). The same gas station operator can provide charging.	Duration of charging being typically longer than with ICE, the parking space may not be sufficient for many vehicles at once. Grid connection can be good or not, depending on the specific location.	Business resistance to introduce a competitor to ICE, for which profit margins might be lower.	Mandate to provide alternative fuels. Bonus-malus tax system, by which gas station without EV charging pay a fee, whose total revenue is (wholly or partially) distributed to newly installed EV chargers at gas stations (fiscal neutrality or positive fiscal flow). ⁴
Workplace, including in industrial areas (reserved area under the control of the employer)	Charging can be integrated in working hours and power loads, with little impact on the grid. Charging can align with the sunny hours of the day. Strong social signal of EV acceptance. Free charging can be part of a company policy for sustainability.	Only the workforce (and possibly the clients and suppliers) can take advantage of the infrastructure. Potentially low utilization until EVs penetrate those groups.	Cultural resistance by employer can prevent deployment. Only relatively larger companies offer parking.	Social kudos to employers offering charging. Tax exemption of free charging (which could have instead been taxed as fringe benefits).
Workplace, including in industrial areas (open area accessible by the general public)	Charging can be integrated in working hours, often including hours of PV production.	At the end of the charge, vehicle owners may be at work without the possibility of recovering the vehicle.	Lack of area managers. Permitting.	Facility management companies and industrial consortia for the area can be engaged and become interested in attracting an operator for the charging station.
Charging point in front of public administration building	High prestige location, with centrality recognition and possible media coverage. If the building is highly attended by the public, good integration with users' habits, PV production, and access to high-traffic areas. Obvious target for public funding.	Possible low utilization rate if the public administration itself does not adopt EVs and EV general penetration is still low.	Cultural resistance and controversy in certain political settings. Lack of engagement by political bodies.	Consensus building across the political spectrum. Free charging offered, at least for a period of time.

⁴ See: ch. 4.4 of Piana et al. (2012), available at: <https://economicswebinstitute.org/policymakers/mitigation%20policies%202012.pdf>

<i>Location of EV charging point</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Shop, commercial high street, shopping centre, supermarket and hypermarket	Highly compatible with shopping habits, so duration of the charging not an issue. Large parking slots already existing. Available for everyone, irrespective of homeownership and employment status. Business interest of the destination to attract clients (especially during take-off EV penetration stage), which could even lead to free charging (compensated by sales in the commercial premise). Charging can align with the PV hourly production (even on-site).	-	Business leniency until mid-to-high EV penetration.	Encourage voluntary installation. Moral suasion with supermarket headquarters. Fiscal temporary incentives. Subsidize installation to support urban high-street for urban renewal.
Charging point in front of a restaurant	Highly compatible duration of charging with food consumption. Pleasant activity during charging. Accessible for people irrespective of e.g. homeownership and employment status. Except for dinner, consistency with PV production timing. Business interest of the destination to attract clients.	-	Business leniency until mid-to-high EV penetration.	Encourage voluntary installation. Make charging part of the restaurant review.
Charging point in front of an automotive dealership	Easy to impose by an EV brand to its franchising partners and dealers. Useful to charge the vehicles shown for sale and test.	It may not be clear if also the previous clients and the public can use it. Logistic access outside business hours may not be guaranteed. Location may be outside high traffic areas and difficult to find.	Dealers may resist electrification, trying to sell (and boast) ICEs instead. Brand-specific ways to engage with the dealers.	Engage with brands and dealers so that they open up the charging opportunity to everyone.

<i>Location of EV charging point</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Charging point in front of a school, university and other educational institutions	Fairly compatible duration of charging with (slow) charging. Outreach to youth in a setting aimed at shaping social norms, favouring conformity to pro-environmental behaviour. Consistency with PV production timing.	Access restricted to students and teachers.	School principals' leniency and bureaucratic barriers.	Pool of teachers as important users, which may be connected with special discounts on quantity purchases supported by the institution.
Charging point in front of elderly homes and other similar social aggregation spaces	Fairly compatible duration of the activity with (slow) charging. Outreach to seniors and special groups.	Access restricted to members.	Leniency by people in charge of the space. Lack of investment capacity and other priorities in investment.	Include EV charging in other pro-sustainability retrofit of spaces, with financial support and sponsoring.
Charging point in front of a cinema, theatre, museum, archaeological site, sport facility, country club, golf club fitness centre, natural part and other (closed or open space) amenity	Highly compatible duration of charging with amenity experience. It promotes visit to worthwhile amenities. Outreach to specific communities, potentially opinion-leaders for their peers. Fair consistency with PV production timing.	Underutilization in the night and seasonally. It may not be interesting for the general public.	Leniency by people in charge of the space.	Involvement of national federation of dedicated organizations and related ministries (e.g. Ministry of Culture, of Tourism, of Sport, etc.). It is both an incentive to visit and to boost pro-environmental credentials of the institution.
Charging point in front of large-scale infrequent event arena (stadiums, concerts...)	Highly compatible duration of charging with amenity experience. Outreach to specific communities. Very large number of cars usually parked there during events.	One would need a lot of chargers during the event, but they would be underutilized in absence of event.	Low profitability of recharging station over the year.	Pop-up movable solutions, with batteries on wheels, to be deployed in large numbers during the event (with people going to the cars wherever they have been parked) and other venues while in the first place there is no activity. For instance, on Sunday during the match the batteries on wheels are at the stadium, during the week they are at theatres and city centres. ⁵

⁵ For a commercial example of this socio-technological approach, see: <https://www.reefilla.com/en/fillee/>

<i>Location of EV charging point</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Charging point in front of a hospital or other health-related premise	High social visibility of charging and of the link across clean air, e-mobility and health. Outreach to health-sensitive social groups. Possible media coverage of the inauguration.	Random relationship between charging time and stay.	Leniency by people in charge of the premise. Insufficient number of charging points leading to depressing experience.	Financial support from institution and sponsors (e.g. a certain recharging operator). Possible pool with institution purchases with doctors, nurses and other employees as important users, which may be connected with special discounts on quantity purchases.
Charging point in front of another typical destination	Reduce the need of on-transit charging (doubling the range for the areas deprived of it). Reduce range anxiety, because of multiple locations. Multiply the actors interested to offer charging (in this case as collateral business). Increase competition, taming excessive oligopolistic pricing.	Logistic access may not be optimal, with preferential or exclusive use by a group of people utilizing the destination. Grid connection to be locally verified.	Leniency by the responsible of the destination who may cognitively ignore the idea of offering charging.	Engaging with collective representatives of the type of destination, to generate interest in the idea of installing across their members. Notify in open data the presence of the charging opportunity, with clear indications if restricted to clients or other groups.

Table 2 – Location of battery swap stations

<i>Location</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
High traffic areas	Density of use opportunity. Fair grid connection (at low rates of charging for the battery to be inserted in the vehicles).	Traffic congestion in the surroundings of the station	Permitting. High capital cost, requiring a captive market of swapping-capable vehicles of relatively large pool.	Explicit time planning for installing stations, with coordinated push for vehicle sales.
Peri-urban areas surrounding large cities	Easy access and exit, with low congestion. Accessible from both the city centre and the surrounding areas.	Possible long detour from typical trips. Not found if not by digital tool suggestion. Risk of low utilization rate.	High capital cost.	Broad communication campaign on the planned swap station.
Main inter-city roads and expressways	Full exploitation of the advantage of quick stop for a long new range. Continuity over several stops for extremely long trips.	Grid connection may not be good. Long trips being a minority of regular trips, there may be seasonal use and	Competition in land use with oligopolistic powers.	Country-wide visibility and awareness of the system. Priority grid access. Require from system operators to provide transparent grid capacity data to the investors.

All this related to a territorial meso-level of areas. If one is now coping with the issue of location efficiency at the very micro-level, one might wonder whether this parking slot allocation is efficient, as the same charger has two outward cables for two incompatible standards (thus can only charge one car). Conversely, one can reflect upon whether this other micro-configuration (in which one charger can dynamically charge four cars) would not be more efficient (in space, service level, connections to the grid) (Figures 3, 4, and 5).

Figure 3
One charger, two cables for two different standards, one car served

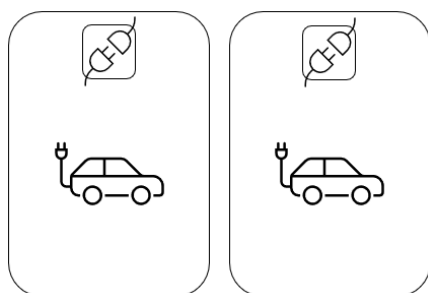


Figure 4
One charger, four cars served

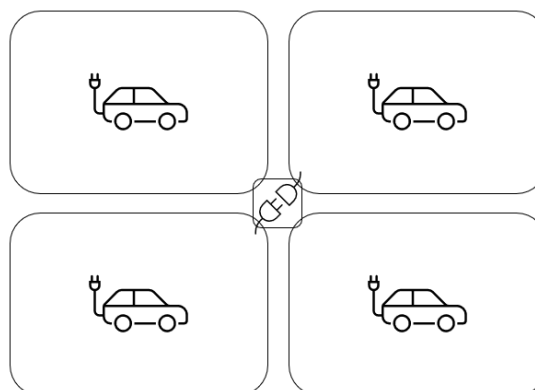


Figure 5
One charger, two cables for two different standards, one car served.⁶



In this report, the focus is not on e-cars alone, but we consider the largest possible definition of EVs, including with a few devoted chapters ("Beyond just cars: exploring e-mobility transportation modes"; "Cultural and regional influences on mobility choices"). In principle, chargers should address the needs of all EVs. In practice, the above Tables 1 and 2 covers e-cars well but less so other types of EVs. Accordingly, additional Tables 3-6 are introduced.

⁶ Photo courtesy of Valentino Piana, used with permission.

Table 3 – Location of charging for E-bikes, e-scooters and other e-MM vehicles (both in ownership and in sharing)

<i>Location</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Home (wall socket and within house and flat removable battery)	Easy and comfortable for the user. Compatible with lifestyles. It pools charging with normal consumption by the household, with no requirement of grid update	Very slow charging.	Awareness at households.	Communication campaigns.
Regional system of swappable charging exchange spots	Peace of mind for long tours (including along pleasant routes). Short time for swapping. Slow and smart charging possible for the battery to be delivered to the vehicle. Powerful motivator for e-bike-based tracks, tourism and inter-city trips, substituting cars if superiority for jam avoidance and parking times is significant.	Infrequently utilized system, with low awareness at policymakers and producers.	Coordination and capital cost. Co-benefit for tourism not internalized. Skepticism by decision-makers.	Scouting for potentially interested investors and operators of the regional system. Involvement of e-bike-sharing companies (for a system that is open also to private customers).

Table 4 – Location of charging for urban e-buses, e-vans and other commercial (last-mile) e-vehicles

<i>Location</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Company depot	Private spaces already used for logistics. They may already have PV systems, thus cutting the operating cost of electricity.	Monopolization and under-utilization of the charging, as it cannot be accessed by other companies, which have to duplicate the point, even if they are nearby.	It increases the financial burden for the company to shift towards EVs.	Subsidise systems located at the depot but open to the public.
City mobility hubs	Devoted areas with multiple services in synergy.	Long occupancy time of charging point keeps it busy and unreliable.	Some hubs may be yet under construction or not conceived with e-mobility in mind. Governance conflicts.	A clear calendar of consultation of all stakeholders to propose changes, integration and further hubs.
Underground parking spaces	Devoted areas without impact on street-level walkability. Possibility of many chargers. Usually, good grid connectivity.	Need to pay for the parking space.	Governance inertia.	Mandatory standards. Gathering data about future needs and promoting the presence of charging proactively.

Table 5 – Location of charging for medium -and heavy-duty e-trucks, e-coaches

<i>Location</i>	<i>Conditions for efficiency</i>	<i>Disturbances to efficiency</i>	<i>Obstacles to deployment</i>	<i>What can be done</i>
Company depot	Private spaces already used for logistics. They may already have PV systems, thus cutting the operating cost of electricity.	Monopolization and under-utilization of the charging, as it cannot be accessed by other companies. If the only available charging, the vehicle range limits total round trip distance.	It increases the financial burden for the company to shift towards EVs.	Subsidize systems located at the depot but opened to other companies. This multiplies the locations for charging.
Multi-brand logistics distribution centres and hubs	Compatible with systemic organization of logistics. Cost-sharing across multiple purchasers. Reduced risk of non-utilization. Visible to non-adopters, providing vivid examples to follow and from which to learn.	-	Logistic layout of the hub to be adjusted. No clear leader in pushing the change. Skepticism by decision-makers.	Map the regional and national hubs, present the technological solutions, call for leaders, induce cooperation, praise announcements and first instances.
Supermarket and warehouses distribution platform	Private spaces already used for logistics. Clear leadership. High power and visibility across thousands of product value chains. Alignment with final consumers request (and supermarket positioning and advertising). Predictable distances covered by vehicles.	-	Leadership inertia or opposition.	Government moral suasion and citizen NGO pressure. Industry discussions.
Highway devoted areas	Essential for mid-range and long-range trips. Compatible with regulatorily mandated breaks for drivers' resting and safety.	Fragmented connection. If only few such stations exist, few fleets will adopt vehicles (with low utilization of the stations themselves). Risk of multiple standards for charging, both at kilowatt levels (e.g. 400kW) and at the megawatt scale.	High investment costs (High-power DC chargers, enhanced cooling systems, reinforced cables, high-capacity grid connections, and often dedicated substations and energy storage systems). Uncertainties about future utilization rates.	Device complete logistics corridors where multiple locations are used by the same vehicles. Promote convergence and compatibility across standards. Supply chain coordination for a critical mass of vehicle purchases with commitment to use the charging stations in the corridors. Vehicles should be able to charge in multiple configurations. Battery swap can be explored.

There are significant synergies across all these locations, because the purchase of e-trucks and e-coaches will be very limited until there are many open locations available. In particular, subsidies to open up company depot locations can unlock the purchase of vehicles with smaller batteries (thus cutting the upfront investment) or much longer utilization of vehicles with the same battery (thus making them much more useful). The addition of charging at regional hubs and along highway corridors can produce the emerging property of a system in which the costs per kilometre are significantly lower than with diesel-fueled vehicles. Logistics is a very cost-sensitive industry, with little operative margins of profitability. Drivers have often difficult working conditions, with little rest, fears from thefts, and may pay the consequences of all traffic jams, construction works and delays with stress, high blood pressure, and obesity. By providing safe and comfortable areas for charging, resting, eating well and healthily, the quality of life of drivers will increase, which in itself is important, favouring thus the appeal of the job and good industrial relationships.

Finally, batteries can power further vehicles, as mentioned above, of which we only signal battery-powered trains for non-electrified routes (leaving to the reader interested in the other types⁷ to adapt these and other reflections made in this chapter *mutatis mutandis*).

Table 6 – Location of charging for battery-powered trains

Location	Conditions for efficiency	Disturbances to efficiency	Obstacles to deployment	What can be done
Train station (platform-side)	Lower cost, flexible installation, possible integration with local renewable energy sources	Time to charge, space in station.	Lack of space, large investment, chicken-and-egg problem (especially if infrastructure and trains are under different companies if they do not cooperate).	Raising awareness of technological opportunities. Setting national standard and goals for (battery-powered) electrification.
Short electrified track sections ("overhead line islands")	More economical, especially in rural or environmentally protected areas, than entirely electrified lines.	Possibly slow speed of trains at the section.	Complexity and cost of infrastructure design, grid integration challenges for the high-power, short duration charging peak loads	International diffusion and standardization

This rather long list of options cannot exhaust further territorial localization of chargers of the many types of EVs that a balanced and sustainable mobility system would require. However, they allow for an initial reference for discussion on "what" to locate "where".

In summary, there is a vast choice of locations for charging, which are not incompatible but rather complementary to each other. On one hand, no area should be without accessibility to some charging. On the other hand, redundancy in charging point in the same location (or nearby) is very important for resilient charging experiences. Data-supported planning using real-time mobility patterns, land use data and grid capacity information, can inform optimal charger placement. There is thus a key need for organizing the multiplicity of locations, to provide certainties to efficient e-mobility developments.

⁷ For instance, the electrification of aviation by electric airplanes, electric vertical take-off and landing aircraft (eVTOLs), and Unmanned Aerial Vehicles (UAVs) is beyond the scope of this study.

Grid infrastructure

Overview of grid infrastructure and EV charging trends

In 2023, home charging accounted for approximately 68% of all EV charging globally,⁸ highlighting the continued dominance of residential solutions. However, the expansion of public charging networks remains critical for mass adoption, particularly for consumers without reliable home charging access. Globally, public charging infrastructure deployment varies significantly:

- Europe has focused on dense urban networks and highway corridors, often supported by European Union mandates.
- The United States infrastructure growth has accelerated under federal and state initiatives such as the National Electric Vehicle Infrastructure (NEVI) programme.
- China leads in total number of chargers installed, supported by aggressive government mandates and integration into new urban developments.

Types of EV chargers and technical characteristics

EVs draw power from the grid in alternating current (AC) form, which must be converted to direct current (DC) to charge the battery.⁹ Technologies such as DC charging stations allow for the current conversion process to occur within the station itself, allowing for larger converters to be used, which provides higher power outputs and shorter charging times.

EV chargers are broadly classified into three levels: Level 1, Level 2, and Level 3. Levels 1 and 2 use AC described above while Level 3 uses DC.

- Level 1 chargers are known for their ease of utilization and low energy intake. This type of chargers uses a standard household 120-volt outlet, found in North America and provides around 8 kilometres per hour of charge, making them suitable for overnight charging, as well as for plug-in hybrid EVs. As other regions, such as Europe, use higher voltage (220V), Level 1 chargers are not a popular option outside of North America.
- Level 2 chargers use higher rates of AC. Level 2 is commonly installed in European households and is also a common choice for offices, parking garages, and public charging stations.
- Unlike Level 1 and 2 charging stations that use steady electricity flows, Level 3 chargers go a step further by utilizing DC with the power output bursting up in the beginning of the charging cycle to significantly reduce charging times, providing a substantial range in a short period and gradually slowing down the volume as it approaches the 100% mark.¹⁰

High Power Chargers (HPCs) typically refer to DC chargers delivering 150 kW or more.

⁸ See: <https://www.iea.org/reports/global-ev-outlook-2024/trends-in-electric-vehicle-charging>

⁹ See: <https://www.climateaction.org/news/ev-charging-the-difference-between-ac-and-dc>

¹⁰ See: <https://www.power-sonic.com/blog/guide-to-level-2-ev-charging/>

Table 7 – Levels of chargers

	<i>Level 1</i>	<i>Level 2</i>	<i>Level 3</i>
Voltage	120V AC	208-240V AC	400-1000V DC
Power Output	1kW	7-19kW	50-350kW
EV Charge Time for a battery of 40kWh (0%-100%)	40-50 hours	4-10 hours	20-60 minutes
Kilometres per hour of charge (with wide variation across vehicles)	3-8 km	16-32 km	290-386 km
Locations commonly installed	Home (predominantly in North America)	Home, workplace, public spaces, public parking	Public charging hubs, highway corridors

Impact on grid

The electricity grid and transportation sector are undergoing rapid, simultaneous transitions to cut emissions. Linked through EV charging, these shifts can both support and strain one another: the grid must provide reliable, affordable power and accessible charging, even as EV charging drives new demand, stresses equipment, necessitates upgrades, affects renewable integration, and creates opportunities for grid services.

Level 1 chargers' slow rates allow most grid systems to accommodate them without major upgrades, especially when distributed across homes.

Level 2 increase localized energy demand and may require upgrades or managed charging solutions to avoid straining neighbourhood transformers during peak periods, if not accompanied by energy efficiency gains and smart devices.

Level 3 chargers place substantial demands on the grid, often requiring dedicated higher-voltage connections, transformer upgrades, and robust infrastructure, especially when deployed in clusters.

Grid solutions to support charging infrastructure

Grid solutions to support charging infrastructure include (but are not limited to):

- Smart Charging Management: Intelligent charging solutions and software distribute energy demand, avoid peak-time overloads, and manage loads dynamically, letting utilities optimize energy use and delay costly infrastructure upgrades.
- Load Forecasting and Planning: Utilities and researchers use modelling and scenario analysis to forecast future EV charging demands and plan investments or upgrades in advance, considering regional and urban/rural grid topologies.
- Grid Topology Adjustments: Urban grids with looped or meshed topologies offer more flexibility and resilience for EV charging, while rural radial systems may require targeted upgrades to handle even moderate additional loads without voltage issues.
- Integration with Renewable Energy: Advanced grid management can coordinate EV charging with renewable generation, battery storage, and V2G capabilities to improve stability and utilize clean energy efficiently.
- Non-Wire Solutions and Local Storage: Utilities are increasingly adopting distributed energy resources and local storage systems to manage increased demand without traditional grid investments, improving resilience in case of outages or peak demand.

Locational fit of chargers

The wide range of locations, described in the chapter on “Location efficiency and its various contributing factors, including the integration with buildings and industries”, have quite natural coupling with one of the three levels of chargers. In general, charging points at the origins of flows (homes, company depots) tend to require Level 1 and 2 chargers, in the assumptions that charging takes place during the night or prolonged stops. Charging points at locations of transit require Level 3 chargers. Destinations, such as workplaces, shopping centres, cinemas, tend to have very predictable available time, so can be provided by equipment at the appropriate level (often for partial charging).

Consumer behaviour and EV chargers

The availability of charging infrastructure remains to be one of the most important variables for consumers when considering purchasing EVs with 29% of surveyed consumers in North America said they would be more likely to switch to an EV if public chargers were as prevalent as gas stations, 24% cited higher charging speeds as a decisive factor, and about 10% stated that a mix of home and public charging access would encourage adoption.¹¹

Prevalence of charging stations is an important factor for consumers' acceptance of EVs and that there are currently not enough charging stations in the United States to soothe the range concerns of average car buyer.¹²

A United States based simulation study using the ParaChoice market analysis model found that public DC fast charging infrastructure significantly outperforms Level 2 charging in promoting EV adoption, increasing electrified mileage, and reducing greenhouse gas emissions. The authors reported that even when DC fast chargers are deployed at only one-tenth the quantity of Level 2 chargers, their impact on EV sales and environmental outcomes remain greater. Model projections indicated that a large-scale national rollout of DC fast chargers could increase 2050 fleet-wide electrified mileage by approximately 8% compared to a baseline scenario without such expansion.¹³

Economic considerations

Charging cost is another key adoption factor. In the United States, refueling a vehicle with gasoline was found to be two times more expensive than charging EVs.¹⁴ Recent surveys, especially by AAA shows that long-term fuel and maintenance savings are among the top reasons consumers consider switching to EVs.¹⁵ The 2025 AAA survey of the United States drivers found that 77% of respondents considering an EV cited gas savings as their primary motivation,

Higher gasoline prices in Europe and recent surveys also confirm substantial long-term fuel savings for EV owners compared to ICE vehicles, and these financial factors are influencing rapid EV adoption across the region. According to a 2025 report from EY and Eurelectric, consumers switching to EVs saved on TCO, with annual savings of 4% in the compact segment, 9% in the family car segment, and

¹¹ See: <https://www.mckinsey.com/features/mckinsey-center-for-future-mobility/our-insights/new-twists-in-the-electric-vehicle-transition-a-consumer-perspective>

¹² How many chargers is enough? Current research and media (Singer, 2016, Mooney, 2016, Davies, 2014, Read, 2013).

¹³ See: <https://www.sciencedirect.com/science/article/abs/pii/S136192091630757X>

¹⁴ See: <https://bcse.org/market-trends/>

¹⁵ See: <https://electriccarsreport.com/2025/06/aaa-survey-americans-slow-to-adopt-electric-vehicles/>

14% for SUVs could be realized on average across key markets in Europe.¹⁶ Adoption of smart charging and technologies such as V2G will widen the gap even further.

These economic benefits are critical for EV adoption, as the lower operating costs offset the generally higher purchase price of EVs. As gas prices fluctuate and electricity rates remain relatively stable, especially for home charging, prospective buyers become more comfortable relying on predictable charging expenses rather than volatile fuel costs. Although some drivers cite upfront costs and concerns about public charger prices, surveys indicate that the long-term savings on fuel and maintenance, EVs cost less to maintain than ICE vehicles, are major motivators to switch to electric, accelerating adoption as financial planning becomes easier and incentives narrow the price gap.

Challenges of scaling up EV charging installations

High installation prices and inadequate government support for EV charging infrastructure remain some of the largest challenges for EV charging networks. The purchase and installation of Level 2 commercial grade charger in the United States costs about USD 6,000 while Level 3, depending on the power output, can exceed USD 100,000 as they require additional labour, more complex installation, and electric utility companies' services (150 kW DC chargers costing around USD 75,000 per charger in the United States).¹⁷

If installation costs are passed on to consumers via higher electricity prices, adoption benefits can be eroded. The ParaChoice study found that once consumer costs increase by only a few cents per kWh for all miles driven, most gains in sales and emissions reductions are lost.

Policy and governmental initiatives

Europe is advancing rapidly in EV charging infrastructure deployment, backed by strong European Union policies like the Alternative Fuels Infrastructure Regulation (AFIR), that requires that fast-charging stations are installed every 60 km along the Trans-European Transport Network (TEN-T) by 2025. Separately, the government of Germany is mulling to require for larger petroleum companies to install at least one EV charger at every gas station.¹⁸ Such mandates, especially when installation process is subsidized can offer additional sources of revenue for gas stations, while decreasing the prevailing range anxiety.

By mid-2025, Europe surpassed one million public chargers, with forecasts requiring roughly 8.8 million by 2030.¹⁹ While public charging networks are expanding with increased fast and ultra-fast chargers, the majority of EV charging happens at private or workplace locations, where smart charging and load management optimize grid capacity and energy use.

The United States Government programme, NEVI Formula Program is a key United States federal initiative funded under the Bipartisan Infrastructure Law with USD 5 billion allocated to states for deploying EV charging infrastructure across the country. Administered by the United States Department of Transportation's Federal Highway Administration (FHWA), NEVI funds primarily target the strategic

¹⁶ See: https://www.ey.com/en_gl/newsroom/2025/03/smart-charging-and-v2g-critical-for-cost-savings-grid-stability-and-renewables-integration

¹⁷ See: <https://www.itskrs.its.dot.gov/2020-sc00472>

¹⁸ See: <https://www.cleanenergywire.org/news/fast-charging-points-become-mandatory-large-petrol-station-operators-germany#:~:text=Large%20petrol%20station%20companies%20will,agreed%20by%20Germany's%20government%20cabinet>

¹⁹ See: <https://www.pilotenergystorage.com/ev-charging-infrastructure-incentives-in-europe-2025/>

build-out of EV charging stations, especially fast chargers, along designated Alternative Fuel Corridors (AFCs) on the national highway system to support long-distance EV travel.

Despite some rollbacks, the current presidential administration is allowing for funding to continue with updated guidance for states to apply for funding with 2025 is on track to be the year for the highest installations of superchargers in the United States.²⁰

In the Philippines, under the Electric Vehicle Industry Development Act (EVIDA), gas stations located within the Comprehensive Roadmap for the Electric Vehicle Industry (CREVI) are mandated to designate dedicated spaces for the installation of commercial use charging stations for the public.²¹

In India, while not a nationwide mandate for existing pumps, the Petroleum Ministry has mandated that all new fuel pumps have infrastructure for at least one alternative fuel, with many opting for EV charging facilities. The Power Ministry has also sought directions to mandate the carving out of space for EV charging stations at existing company-owned and company-operated (COCO) fuel pumps.²²

Technical and integration challenges

Lack of integration between types of connectors used by different auto manufacturers results in multiple plug standards across automakers, which can frustrate users and slow EV adoption. Industry cooperation on standardization is essential.

High-power charging can strain local grids, requiring investment in distribution infrastructure.

Scaling EV charging infrastructure is not merely a technical task as it is an economic, behavioural, and policy challenge. While DC fast chargers deliver the greatest impact on adoption and emissions reduction, their benefits depend on careful management of deployment costs and electricity pricing strategies. Government actions, such as policy mandates coupled with subsidies, can greatly accelerate deployment, while industry-wide collaboration on charging network standardization and technical requirements, such as uniform EV charging ports, can further increase adoption and deliver additional financial returns. Emerging technologies like V2G offer the potential to integrate charging infrastructure into the broader energy system, enhancing energy security and energy resilience while creating new revenue streams for stakeholders.

Charging infrastructure and energy storage

The relationships between charging infrastructure and energy storage have three main venues of discussion:

1. Battery-buffered charging infrastructure, thanks to which there is the decoupling of the moment of vehicle charging, dictated by users' needs and accessibility, from the moment of grid loading, which can be optimized to keep cost and carbon intensity into account. The intermediation of batteries may be expensive and not always necessary; however, from a systemic point of view, it is advisable to explore ways to have it implemented in certain locations (e.g. where the grid is

²⁰ See: <https://insideevs.com/news/767059/record-ev-charger-deployment-q2-2025/>

²¹ See: <https://cleanairasia.org/sites/default/files/2025-07/COMPREHENSIVE%20ROADMAP%20FOR%20THE%20ELECTRIC%20VEHICLE%20INDUSTRY%20%5B%202023%20version%20submitted%20to%20DOE%5D.pdf>

²² See: <https://energy.economictimes.indiatimes.com/news/power/fuel-stations-asked-to-have-at-least-one-ev-charging-station/77988741>

particularly weak or peak use is likely to be particularly out-of-sync with renewable production) or more extensively.

2. Energy storage contained in EVs, especially with the trend of larger and larger batteries, way exceeding the needs of daily travelling, and with the possibility of devoting fleets of batteries-on-wheels for coping with anticipated grid events;
3. Good location of EV charging with respect to grid needs, including distances to sub-stations and current levels of grid congestion.

These discussions are not detached from the broader issue of storage in support of smart and general grid, independent on mobility. In this context, a niche of growing importance is the opportunities opened by giving a second life to vehicle batteries.

The waste of discarded EV batteries is a frequent criticism of electrification. Second-life batteries (SLBs) offer a practical way to mitigate these risks by repurposing retired EV packs that retain about 70-80% of their original capacity (Atlas Policy 2025). While this is insufficient to power a vehicle, it remains valuable for less energy-intensive applications.

Because SLBs are substantially cheaper, often costing only 25-40% of a new battery, they present a cost-effective alternative for many use cases. Off-grid charging in remote or rural areas, backup power during natural disasters, and emergency roadside assistance are among the most prominent applications. When paired with advanced battery management systems, SLBs not only extend their lifespan but also ensure safety, reliability, and optimized performance.

By reducing costs, alleviating supply chain pressures, and supporting sustainability goals, SLBs provide both economic and environmental benefits while advancing EV charging accessibility. However, challenges exist as SLBs face hurdles such as complex and costly safety certification requirements, variability across EV battery models that complicates refurbishment, and uncertainties in long-term performance due to the lack of standardized health assessments. These factors add expense and time, potentially slowing widespread adoption.

Conversely, a policy in which the battery cost at the moment of EV purchase is partially born by an energy operator, gaining ownership and use of it after a predetermined number of years (and/or kilometres made), could reduce the upfront cost of the EV below ICE alternatives and assure quality and compatibility of batteries for later use in the energy system.

Vehicle-to-grid integration

Beyond meeting current demand, EV charging infrastructure can evolve to support grid resilience through innovative technologies, such as the V2G. As demand growth for electricity is projected to increase by 3% annually through 2035, EVs have the potential to not only consume, but to distribute electricity using the V2G technology.²³ This bidirectional technology is currently being piloted across the world, including in Europe, United States, and Japan.

Europe is making varied strides in rolling out V2G technology, with notable progress in France, the UK, the Netherlands, and Germany. In France, owners of the Renault 5 are able to use bidirectional AC charging via the PowerBox Verso and a dedicated energy contract, supported by a favourable regulatory environment. In the UK, several pilot projects exist, and while regulatory and technical hurdles exist, including high approval costs for chargers above 3.5 kW, OFGEM is advancing a more unified V2G

²³ See: <https://www.iea.org/energy-system/electricity>

framework; Nissan plans to introduce affordable V2G technology starting in 2026, with broader V2G adoption expected from 2025 onward. The Netherlands leads with numerous city-level pilots, such as Hyundai's V2G trial in Utrecht, but full commercial deployment is still in development. Germany shows significant technical readiness, with demonstrated communication systems and grid services potential, yet widespread deployment is hampered by a slow smart-meter rollout (fewer than 10% installed) and a fragmented network of DSOs complicating standardization and regulatory consistency.²⁴

Across the United States, V2G programmes are rapidly expanding, with state-run pilot initiatives, utility incentives, and federal support driving technology deployment, market development, and research for EV grid integration. For example, the State of Massachusetts launched a pilot V2G programme in early 2025.²⁵ This two-year initiative is deploying 100 bidirectional chargers across residential homes, commercial fleets, and school districts, targeting a diverse range of settings, including an emphasis on equity by prioritizing disadvantaged communities and municipal fleets. The programme offers bidirectional chargers at no cost to participants, enabling eligible EVs to act as mobile energy storage units. By the end of 2025, Massachusetts expects to have added up to 1.5 MW of flexible, distributed energy storage to the grid directly through these EV chargers.

A growing trend in the United States is the rollout of electric school bus V2G pilot projects, now spanning at least 19 states and involving 26 utilities.²⁶ These initiatives tap into bus batteries to provide grid support and backup power for critical facilities during outages. For school districts, they create new revenue streams while also showcasing the ability of buses to deliver multi-hour grid services.

Using EVs as a mobile battery storage system provides various benefits for countries, communities, and individual EV owners. Being able to use EVs as alternative power generators can enhance countries' responses to national emergencies such as blackouts. Furthermore, electricity rates that vary by time of day could provide a financial incentive for EV owners, offering ability to capitalize on volatilities in electricity price and demand.

The main challenge for technology remains the aging grids that do not have the capacity to offtake electricity. Other concerns include potential impacts on EV battery lifecycle.

Telecommuting, digital infrastructure, and smart cities

Telecommuting represents one of the most significant paradigm shifts in mobility and urban development in recent history. Accelerated by the COVID-19 pandemic, remote work has transitioned from a temporary contingency to a defining feature of the modern labour landscape. In 2020, approximately 42% of the United States workforce was engaged in full-time remote work, with future projections estimating that nearly a third of the workforce will continue working remotely on a permanent basis (Baguma, 2023). This transition is not only altering how people work but also reshaping the infrastructure, mobility patterns and energy demands of urban environments.

At the core of this transformation lies digital infrastructure. "Digitalization is already improving the safety, productivity, accessibility and sustainability of energy systems. But digitalization is also raising new security and privacy risks. It is also changing markets, businesses and employment" (IEA, 2017). High-speed broadband, cloud computing, cybersecurity frameworks and emerging 5G networks now constitute the essential foundation for distributed work. These technologies enable real-time

²⁴ See: https://mobilityhouse-energy.com/int_en/knowledge-center/article/v2g-progress-in-each-country

²⁵ See: <https://pv-magazine-usa.com/2025/02/28/massachusetts-will-pilot-vehicle-grid-integration-statewide/>

²⁶ See: <https://www.wri.org/update/electric-school-bus-v2g-lessons-examples>

collaboration, decentralized productivity and seamless data access, thus reducing dependence on traditional office spaces and daily commuting. The effects are tangible: a 30% reduction in rush hour congestion in many cities has alleviated pressure on transportation networks, while public transit systems face both declining ridership and opportunities for strategic reinvention.

Telecommuting has also transformed land use and housing markets. With fewer geographical constraints, millions of individuals are opting to relocate to suburban and rural areas, increasing housing demand outside urban cores. This decentralization presents both risks of urban sprawl and prospects for revitalized, human-centered planning. The “15-minute city” model, where daily needs are accessible within short walking or cycling distances, has gained traction as remote work blurs the lines between home, work and community.

From an environmental perspective, the sustainability benefits of telecommuting are significant. Remote work has contributed to a reported 54% reduction in emissions from daily commuting and a 30% decline in office energy consumption (Baguma, 2023). Additionally, digital workflows have reduced paper use and business travel, advancing broader climate and resource efficiency goals. Distributed work arrangements also align with smart grid systems and renewable energy integration, as staggered energy demand supports a more balanced and resilient energy ecosystem.

Nonetheless, the telecommuting transition is not without its challenges. The digital divide—defined by disparities in equipment access, internet connectivity and digital literacy—threatens to reinforce existing socioeconomic inequalities. Over 21 million Americans still lack reliable broadband, and low-income and rural communities often struggle with inadequate digital infrastructure. Addressing these disparities is imperative to ensure an equitable transition and to avoid excluding vulnerable populations from digital economic opportunities.

Looking forward, infrastructure investment needs to prioritize universal connectivity and adaptable work environments. Public investments in broadband—such as the proposed USD 100 billion in nationwide digital infrastructure—represent a critical step. Meanwhile, immersive collaboration technologies (e.g., virtual and augmented reality) and AI integration offer new frontiers in virtual presence and productivity.

To support sustainable mobility goals, cities need to treat digital infrastructure as a core component of public service – equivalent in importance to water, electricity or transport. Integrating digital planning into urban design, balancing virtual and physical community needs, and ensuring inclusive access will be central to realizing the long-term benefits of telecommuting. As cities adapt, the strategic alignment of location efficiency, digital connectivity and e-mobility will shape resilient, inclusive and low-carbon futures.

In parallel, the development of Smart cities is broader than mobility but has profound implications for mobility. The presence of sensors on roads and other places provides input for models and elaborations, including supported by AI (artificial intelligence), which then activate “actuators”, be they physical, legal, digital or behavioural. For instance, overcoming threshold of air quality deterioration can lead to identify areas of the city where polluting vehicles are restricted (and building and industries restricted in their polluting outflows). The multiplication and specification of new actuators is the key challenge for the Smart City, since sensors have become cheaper, more reliable, easier to deploy and put into networks, the Internet of Things (IoT) is largely operational (although not yet deployed at scale) while the elaboration intelligence is also making quick improvements and radical transformations. Actuators that are effective, legal, socially acceptable and easy to implement are still often lacking.

GOVERNANCE

Defining the role of governance

The role of governance in facilitating the e-mobility roll-out and location efficient solutions is oriented, with varying degrees, towards performance and processes. A mode of governance that emphasizes performance focuses on effectiveness, favouring a faster, less bureaucratic, and more pragmatic approach to decision-making and implementation of policy measures and regulatory actions. This performance-oriented approach can improve the e-mobility transition by enabling quicker deployment of charging infrastructure, reduced bureaucratic hurdles, and faster adaptation to technological innovation.

A more process-oriented notion of governance, on the other hand, highlights the need for greater collaboration, co-creation, and active engagement of civil society, citizens, and other diverse stakeholders. Particularly in urban areas with spatial confinements and where space is contested, the involvement of civil society becomes particularly relevant. This process-oriented approach is more communicative and collaborative across sectors and governance-levels. This mode of governance can improve the e-mobility transition through a greater degree of legitimacy, consensus, and more thorough planning.

These two governance approaches may stand in conflict (Graaf and Paanakker, 2014) and must therefore be balanced. This duality in governance approaches is captured by the definition that governing “[g]overning can be considered as the totality of interactions, in which public as well as private actors participate, aimed at solving societal problems” (Kooiman, 2003).

To ensure justice in the e-mobility transition, governance of the transition also must consider different notions of justice: distributive, procedural, and recognition justice.

- Distributive justice – the distribution of benefits and burdens across a society, e.g., equitable access to charging infrastructure; avoiding charger clustering in affluent areas and ensuring access in underserved areas
- Procedural justice – the participation and inclusion of relevant actors, e.g., involving (marginalized) groups in location planning
- Recognition justice – the acknowledgment of diverse (mobility) needs, values, and (rural/urban) practices, e.g., providing four-wheeled e-scooters for elderly people (Israel et al., 2024).

Considering these forms of justice ensures that the mobility transition is not only fast and environmentally effective but deliberative and socially equitable.

Governance models for e-mobility: the role of public participation

A prominent part of a more process-oriented mode of governance constitutes collaboration and co-creation. Co-creation can be considered a “collaborative and participatory process in which multiple stakeholders – ranging from public administrations to citizens, businesses, academia and civil society – work together in designing, implementing and evaluating public services and policies” (Müller et al., 2025). Co-creation can make decision-making and planning processes more democratic, fair, and legitimate (Wippoo & Kortlander, 2021).

One group that is sometimes least considered in a transition are citizens. Since citizens—whether they own an EV or not—are directly and indirectly affected by the e-mobility transition and often have limited ability to influence policy mixes (Transport Studies Unit et al., w.d), there is a need to involve them in

decision-making and planning processes. A 2019 survey by the Bertelsmann Foundation in Germany showed that a majority of respondents stated that they did not feel sufficiently involved in local decision-making. In large cities, the proportion of dissatisfied respondents was almost two thirds (Gehne et al., 2019). Involvement can range from sharing information to enabling co-creation. Including citizens can increase their acceptance towards political interventions, help include local knowledge to address unknown-unknowns that are relevant for effective location efficient EV-solutions, improve participants' understanding, and fostering the intrinsic democratic value of inclusion.

The following represents three few notable examples that indicate how successful collaboration in the e-mobility transition can look like:

1. A notable example of co-creation is the arrangement of EV-relevant policies in Bristol. The city strongly enabled participation in the design and delivery of policy instruments relevant to the electrification of modes of passenger mobility (Transport Studies Unit et al., 2024). Participatory activities included online consultations, a Citizen Assembly, co-production, and public consultation, including 'targeting hard to reach groups', feedback to business engagement team, and protests by groups/residents with forms of disability (Transport Studies Unit et al., 2024). Meaningful forms of participation can increase the satisfaction with political authorities and legitimacy (Escher and Rottinghaus, 2023).
2. Another noteworthy example represents the DYN@MO project, which is part of the European Union co-funded CIVITAS initiative, that assists cities to implement sustainable transport solutions. Four European cities – Aachen (Germany), Gdynia (Poland), Koprivnica (Croatia) and Palma (Spain) support the implementation of Sustainable Urban Mobility Plans (SUMP) (CIVITAS, 2014). One of the project's key lessons is that diverse stakeholders should be involved in an early stage of a process, ideally already in the vision-building process (Langweg, 2014). Deploying both traditional and online participation formats, several stakeholders and experts as well more than 500 citizens were involved in the creation of the mobility plan for the year 2050 in Aachen and engaging with thematic experts (Langweg, 2014). Clear and understandable language and avoidance of scientific jargon was found to be a critical component of communication between experts and lay people (Langweg, 2014). Making use of digital participation formats was found to be important in the project's cities, since these digital formats attract younger demographics better than traditional formats, e.g., consultation or workshops (Romanowska & Sancho, 2014). Addressing wider audiences is critical for creating a common vision and ensuring greater legitimacy.
3. Another example of public participation for e-mobility comes from the North Sea project "North Sea Electric Mobility Network", or "E-mobility NSR", in short. Aiming to improve awareness of e-mobility, the project launched a physical, mobile E-Mobility Information Centre (EMIC) that provided general E-mobility information to private and public stakeholders and access to a virtual EMIC (Kath, 2014). The project was represented by experts that answered questions and offered test-drives in an EV. According to the project, these EMICs have the potential to be vital for generating awareness, disseminating information, and learning about sustainable travel behaviour (Kath, 2014).

Finally, engaging diverse stakeholders in a meaningful establishes trust between stakeholders, thereby having the potential to make implementation smoother. Building trust, however, requires investing time and effort (Wippoo & Kortlander, 2021). Also, trust can be breached if participants do not feel like they are being taken seriously, e.g., if their recommendations are not being implemented. Thus, it is

important to demonstrate how the results of participation process are taken up (Wippoo & Kortlander, 2021).

In sum, meaningful co-creation and participatory processes can increase acceptance of policies, improve understanding of complex issues, and enhance the perceived legitimacy of decision-makers. Early and inclusive engagement of diverse stakeholders, using accessible and clear communication, is crucial for fostering trust and collaboration. While such processes require additional time and resources, thoughtful design that demonstrates how participant input is considered can support more democratic and effective implementation of e-mobility solutions.

Promoting e-mobility: awareness and engagement strategies

Effective communication, outreach, and engagement with diverse audiences are critical to prevent gaps in essential information and awareness, and to ensure a widespread adoption of EVs. For example, consumers may not be aware of readily available EV incentives, discounts on public charging stations by electric utilities, or may be unfamiliar with the charging practices altogether (Forthmobility, 2024). Thus, there is a need for clearly communicating the incentives and changes in social practices to ensure an effective e-mobility rollout.

A multifaceted outreach and communication strategy combining outreach, communication, and engagement is likely to be more effective than relying on a single approach. While outreach describes a bidirectional and interactive way of engagement with a specific audience, communication refers to a one-way engagement between the sender and receiver (European Commission, 2020).

Communication can be viewed more nuancedly by the frames and messages. Frames emphasise certain aspects in communication. Three different frames (Lindenberg & Steg, 2007) can be identified that are relevant to e-mobility:

- gain frame ("what's in it for me?")
- emotional frame ("to feel better right now")
- normative frame ("acting appropriately")

A study by Decrinis et al. (2023) tested the effects of these three frames on EV choices of 170 employees of a German automotive company. The study found that the gain frame – applied in email exchange – exhibited the strongest and most durable effect on electric car choices. Emphasizing cost savings worked better than highlighting sustainability as a salient corporate value. Nonetheless, normative messages were still found to have positive reactions if applied at an early stage in the decision-making process before intention formation takes place. The authors suggest that emotional frame should be applied in the contexts where employees tend to act quickly and affectively upon a prompt (Decrinis et al., 2023).

The UK Government conducted a behavioural science trial to explore how messaging could encourage drivers to consider EVs when renewing their car tax online. The trial, run by the Department for Transport (DfT), the Driver and Vehicle Licensing Agency (DVLA), and the Cabinet Office, tested eight different messages over a period of 130 days. The messages differed in the framing, highlighting cost savings, environmental benefits, social norms, and policy developments related to the transition to EVs. The goal was to identify which types of messaging most effectively prompted individuals to visit the Go Ultra Low (GUL) website, a platform promoting EV adoption (UK Government, 2021).

The results showed that the most effective message referenced government policy: *"The Government are consulting on ending the sale of new petrol, diesel and hybrid cars and vans by 2035 or earlier. Are you ready? Make your next car electric"*, framing EV adoption as a timely and relevant choice. Another

highly effective message leveraged social norms, noting the growing number of drivers switching to electric cars each month. Overall, the trial demonstrated that messages emphasizing policy developments and social trends were more persuasive than generic informational messages, providing insights for future communications aimed at accelerating EV uptake (UK Government, 2021).

What can also advance awareness around and adoption of e-mobility is when communication and campaigns by municipalities and Member states more clearly explain how consumers can play a role in a decentralized energy system and how they can financially benefit from integrating EVs to home energy use (Kotter et al., 2020). Additionally, sharing successful projects and encouraging consumers to participate may help to replicate those successes elsewhere (Kotter et al., 2020).

Moreover, it is helpful to tailor recommendations specifically to the desired EVs. Partners of the Baltic Sea Region (BSR) Electric project have proposed communication and public awareness recommendations for several EVs (e-vans, e-buses, e-bikes, e-scooters, e-ferries) (BSR, 2018). These recommendations include promoting positive overall life- and health-related benefits (E-busses), pointing out the innovativeness (E-ferries), enabling opportunities for people for try-outs (E-bikes), creating a modal shift (using an e-bike as an alternative to motorized means of transport) (E-bikes), and creating local business ambassadors out of first movers (E-Vans and e-logistics) etc. (Kotter et al., 2020). Moreover, debates, rallies, public events may help to promote e-cars.

In all this, it is important to communicate and highlight “the intrinsic value of sustainable actions, such as personal health benefits, community impact, or alignment with values”, as more widely discussed in UNECE (2025). This is true both for individuals and for companies: “Internal economic logic is the strongest trigger: companies act more ambitiously when decarbonization aligns with their strategic interests. Positive motivators outperform external drivers in both influence and impact on ambition levels” (Buettner et al, 2025).

Distinguishing rural and urban mobility contexts

The mobility needs of urban and rural areas across Europe present markedly different challenges, shaped by population density, infrastructure availability, economic activity, and spatial organization. Understanding and addressing these unique mobility landscapes is essential for developing sustainable, inclusive, and efficient transportation systems that contribute to both national decarbonization goals and local development.

Key differences in rural and urban mobility

In rural areas, residents typically travel longer distances to access essential services such as healthcare, education, and employment. These trips are often undertaken using private vehicles due to the limited availability and coverage of PT options. In many regions, the absence of adequate pedestrian or cycling infrastructure further entrenches car dependency, exacerbating social exclusion and increasing the cost of living.

Conversely, urban areas experience shorter, more frequent trips. Public transport systems are often more developed and heavily utilized. Additionally, cities tend to have more active transport infrastructure, such as bicycle lanes and pedestrian zones. However, urban environments also face acute issues such as traffic congestion, high emissions, and constrained physical space for mobility expansion. These challenges disproportionately affect vulnerable populations, including those with limited financial resources, older adults, and persons with disabilities.

Barriers to sustainable mobility

Mobility in rural areas is frequently constrained by infrastructure gaps, especially in terms of PT coverage, digital connectivity, and charging stations for EVs. These limitations hinder the deployment of modern mobility solutions and restrict residents' access to opportunities.

Urban areas, on the other hand, struggle with overcrowded transit systems, parking shortages, and the negative externalities of combustion engine vehicles. While urban residents may have more mobility choices, these options are not always equitably distributed, particularly in informal settlements or lower-income neighbourhoods.

The role of E-Mobility

The rapid evolution of electric mobility technologies offers promising pathways to address both rural and urban mobility issues. In rural areas, EVs can help provide last-mile connectivity, especially when integrated into shared mobility or community transport schemes. However, the success of such interventions depends on the availability of charging infrastructure, affordability, and public awareness.

In urban contexts, e-mobility contributes to reducing air pollution and congestion, particularly when integrated with PT and active mobility networks. The deployment of MM solutions such as e-bikes and e-scooters also complements traditional transit options, enhancing first- and last-mile connections.

Charging infrastructure development remains a cornerstone of effective e-mobility deployment in both settings. Rural areas require targeted investments and innovative solutions, such as mobile charging units or solar-powered stations, to address geographic spread and lower usage density. Urban areas need smart, grid-integrated solutions capable of handling high demand and ensuring equitable access. In terms of governance, it's important to give weight to both urban and rural actors, which can be quite different. There are established channels for such cooperation. In addition, involving and establishing rural and urban Living Labs can be an appropriate and effective way to solicit expressions of interests, open communication, testing of solutions.

Policy strategies and infrastructure needs

Rural-focused policy actions

- Expand PT coverage and introduce demand-responsive transport services.
- Facilitate carpooling across people living in the same area;
- Provide financial incentives and subsidies for EV ownership and shared transport options
- Remove obstacles to sharing mobility in rural areas, where usually the density is low and is difficult to remunerate it, thus rural municipalities could be helped in purchasing vehicles and put them at the disposal of their inhabitants at differentiated fees (e.g. free for certain amounts of use or certain groups)
- Support multi-modal hubs and charging station networks in small towns and remote regions.

Urban-focused policy actions

- Introduce congestion pricing and low-emission zones to reduce urban vehicle traffic.
- Enhance investment in mass rapid transit systems (including bus rapid transport), walking and cycling infrastructure.
- Promote electrification of public and freight fleets to achieve air quality targets.

Integrated and inclusive planning

A holistic approach is essential to bridge rural-urban mobility divides. This involves:

- Connecting rural and urban areas through efficient regional transport corridors.

- Aligning land-use and transport planning to reduce travel distances.
- Assuring universal accessibility, particularly for vulnerable and disadvantaged populations.
- Equitable access to mobility must be a guiding principle in all policy formulations. Transport systems should be designed to meet the needs of all users, irrespective of income level, physical ability, or geographic location. Digital tools, such as MaaS platforms, can help tailor user experiences and facilitate access to integrated transport modes.

Since equitable access to affordable electric mobility and charging infrastructure remains a key challenge in urban and particularly rural areas, it is important that this aspect is reflected in national and municipal policy formulations.

In synthesis, distinguishing the mobility contexts of rural and urban areas is a prerequisite for location-efficient and climate-resilient transport systems. As Europe as a continent moves toward carbon neutrality, recognizing and addressing the nuanced challenges of each context will enable more effective and inclusive mobility planning. E-mobility, supported by robust infrastructure and integrated policy frameworks, holds transformative potential to close mobility gaps, reduce emissions, and improve quality of life across both rural and urban landscapes.

Between e-mobility and immobility: exploring transport poverty

To arrive at an explanation of how the concept of transport poverty can be applied, a suitable starting point might be to begin with the concept of poverty. Poverty can be described as a state when an individual's or a group of individuals' resources, especially physical resources, are insufficient to meet their basic needs. Living in poverty means having significantly fewer resources than what is necessary to meet one's minimum needs, including participating in social life. Poverty is characterized by a pronounced lack of well-being, low incomes, and the inability to acquire the basic goods and services required for a dignified life. It is also associated with "low levels of health and education, poor access to clean water and sanitation, inadequate physical security, lack of (political) voice, and insufficient capacity and opportunity to improve one's life".

Access to affordable, safe, and reliable transportation can play a significant role in developing or breaking poverty – both in the short term, for example, the ability to educate oneself and seek employment, make purchases (e.g., find offers at lower prices), and access necessary services and care; and in the long term, for the ability to participate in social life at large and exercise one's democratic rights. The broader concept of social exclusion is also used to describe participation in social life. The concept of social exclusion can, but does not have to, include poverty. According to (Levitas 2007) cited in (Lucas, 2012), social exclusion is "the lack of or denial of resources, rights, goods, and services, and the inability to participate in normal relationships and activities available to the majority of people in a society, whether in economic, social, cultural, or political arenas."

If access to transportation plays a role in social exclusion, it is referred to as transport-related social exclusion. Lucas (2012) cites the following definition of transport-related social exclusion: "The process by which people are prevented from participating in the economic, political, and social life of society due to reduced access to opportunities, services, and social networks, partly or wholly due to inadequate mobility in a society and an environment that assumes high mobility".

Travel and transportation are often viewed from a socio-economic perspective as a "necessary evil" – a cost – for us to meet, exchange services, work and earn money, educate ourselves, take care of our health, partake in culture, or conduct other transactions with each other, whether between individuals, companies, organizations, or authorities. To describe the possibility of transactions given this

resistance, we use the concept of accessibility, which can be defined as: "the opportunities to participate in social life by overcoming obstacles (e.g., a distance – often thought of as a physical distance, but for accessing services via other communication services, e.g., on the internet, one thinks more about time consumption and other resistances, such as digital knowledge, etc., that prevent us from doing what we actually want to do)." – that is, the potential or benefit of, for example, being able to access an offer that matches the individual's needs and desires, to offer one's services (labour, trade, etc.), or to acquire knowledge, professional networking, or purely social exchange.

Accessibility increases with the extent of "social life" and decreases with the resistance or distance to reach it. The definition also considers that activities can be accessed, and services offered by means other than through the transport system. The definition identified three main components of accessibility: population/individual/demand, supply/society, and various obstacles. Deficiencies in accessibility can lie in any of these components or in unfavourable combinations of them. Problems arise when deficiencies have serious consequences and, for one reason or another, cannot be addressed by whoever is responsible for them.

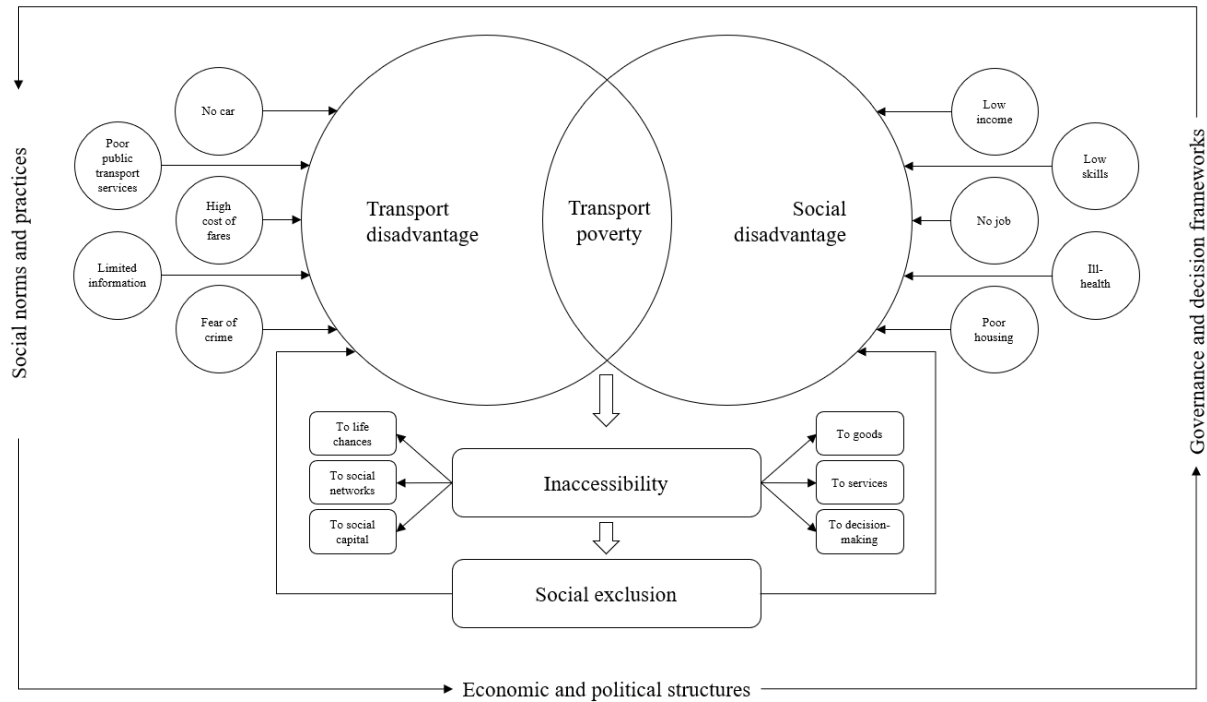
Lucas (2012) writes about "transport disadvantage" and its connection to social exclusion. Lucas argues that by placing "transport disadvantages" in a perspective of social exclusion, it helps policymakers understand that: the problem is multidimensional, i.e., it can be attributed to both the individual's circumstances and the institutions and structures in the surrounding society; it is relational, i.e., an unfavourable circumstance (disadvantage) is seen in direct comparison with normal human relationships and activities in the rest of the population; it is dynamic in nature, i.e., it changes over time and space, as well as during the lifetime of the person affected.

How Lucas views the connections and feedback loops between unfavourable transport conditions, transport poverty, social vulnerability, lack of accessibility, and social exclusion is shown in Figure 6.²⁷ Around these processes, economic and political structures, governance and decision-making regulations, as well as social norms and practices, govern. An interpretation of Figure 6 is that unfavourable transport conditions and social vulnerability interact to create what we call transport poverty, which is a situation of persistent lack of accessibility that can lead to social exclusion. The surrounding structures can reinforce or counteract these processes.

Figure 6

Connections and feedback loops between unfavourable transport conditions, transport poverty, social vulnerability, lack of accessibility, and social exclusion (including overarching frameworks, norms, and governance structures)

²⁷ K. Lucas. Transport and social exclusion: Where are we now? Transport Policy, Vol. 20, 2012, pages 105-113, DOI : <https://doi.org/10.1016/j.tranpol.2012.01.013>.



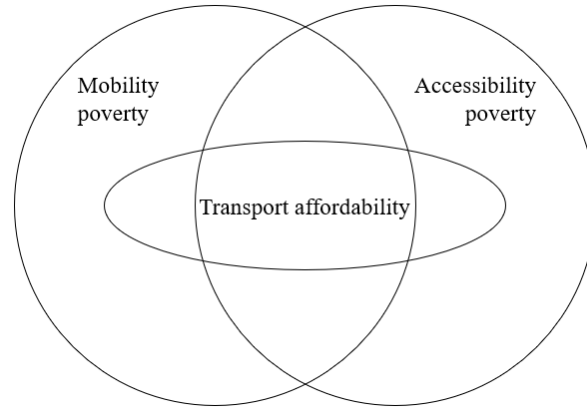
Source: Adapted from: K. Lucas. Transport and social exclusion: Where are we now? *Transport Policy*, Vol. 20, 2012, pages 105-113, <https://doi.org/10.1016/j.tranpol.2012.01.013>.

According to (Lucas, 2016), there are two reasons why it is not easy to define transport poverty briefly, solely based on the household's unmet needs. The first reason is the fact that transport needs are largely individual, and therefore transport poverty should also be perceived individually. The second reason is that the benefit of mobility and accessibility is indirect to the actual benefit, which is to partake in activities, offerings, and services, or to offer one's labour, at the destinations. These activities depend greatly on the context, socially, geographically, and temporally. It is therefore unclear whether "transport poverty" should refer to a lack of transport supply, basic mobility, or accessibility to goods, services, or daily activities.

Lucas (2016), clarifying how the concept of transport poverty is to be used, argues that there are different aspects of the concept that are overlapping but also partially independent of each other (Figure 7). These aspects can be referred to as lack of mobility (availability), accessibility, affordability, and exposure to externalities:

Figure 7
Transport poverty and related sub-concepts

- Mobility availability. A systematic lack of transport options (often motorized), adapted to the individual needs, creating difficulties in moving, linked to a lack of services or infrastructure.
- Accessibility. Difficulties in reaching important destinations and relevant services (employment, education, healthcare, shops, etc.) of sufficient quality and within reasonable distance, travel time, and cost, with available modes of transport.
- Economic affordability. Transport options are too costly for a household to access essential services and activities and participate in social life.
- Exposure to externalities. A disproportionate exposure to the negative externalities of transport, such as traffic injuries and air pollution.



Source: Adapted from: K. Lucas et al. Transport poverty and its adverse social consequences. Proceedings of the Institution of Civil Engineers - Transport 169(6):353-365, 2016, DOI: [10.1680/jtran.15.00073](https://doi.org/10.1680/jtran.15.00073)

Other aspects significant for transport poverty are the sense of safety and security. In socially vulnerable areas, for example, families (especially fathers) drive their children more frequently, prohibit children from going out at certain times of the day, thus limiting their lives and increasing their car dependency. Women avoid traveling by PT, walking, or cycling to a greater extent than men due to a lack of a sense of safety. Note that all the first three aspects above – mobility, "accessibility" (in a narrow sense), and economic affordability – can be included in our more general concept of accessibility. The fourth, exposure to externalities, reveals more of a democratic deficit – a lack of connection to, and participation in, the decisions that shape society.

A variant of transport poverty can arise due to excessive mobility or dependence on mobility. This occurs when transport needs and transport costs become so large that they encroach on the household's other needs, without the household being able to escape its situation. This is often referred to as fuel poverty, which mainly affects rural areas and smaller towns, and forced car ownership, which can also affect metropolitan areas with sprawling suburban development (urban sprawl). They can be considered special cases of transport poverty, as they also have both subjective and structural causes. They are, of course, linked to access to services, shopping, and workplaces (location), a basic PT supply, and planning and construction of infrastructure, e.g., for cycling. If there are alternatives, car ownership is not "forced" in the same way, and high fuel prices are not as noticeable. But they also have subjective causes, as people perceive to varying degrees what options they have.

BEHAVIOUR

This chapter addresses key aspects of spontaneous and policy-induced behaviours of households for private mobility: purchasing, leasing, renting, sharing, using vehicles and accessing mobility services. It recognizes the fundamental role of households' choices, preferences and perceptions (e.g. of risks and opportunities), constraints and variety. It enlists a wide range of policy measures that can be taken to stimulate certain behaviours and make the adoption of system-efficient e-mobility easier and faster.

It covers behavioural rules for EV charging (location and timing), since it would not be systemically efficient to have a lot of installed charging station without utilization for long periods of time.

Across its sub-chapters, it reveals the depth of impacts that people's behaviour has in influencing the efficiency of e-mobility location and in selecting venues to reduced mobility needs, thus lower emissions and energy consumption.

Incentivizing EV adoption

The purchase, leasing, rental and sharing of EV can be incentivized by several policy approaches, including both monetary and non-monetary measures, with tax and subsidies being examples of the former, communication and experience exchange platforms for the latter.

After a solid yet concise introduction of the diffusion theory and some insights from a few countries, the sub-chapter makes a relatively long list of measures, related to different stages of the diffusion process of EVs, which could be undertaken, after more detailed analysis of the country conditions and with further design refinement, by public authorities and stakeholders. The general logics is not prescriptive but aimed at offering a menu to policymakers.

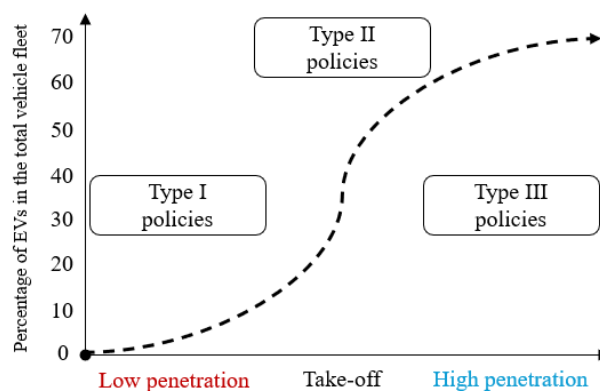
When new technologies are introduced, different segments of society often adopt innovations at varying speeds or levels of interest. There are several approaches to analyzing the introduction of new technology and its market penetration. One approach is to examine the product's rate of introduction and its market share, and to divide the product's adoption process into different phases based on these factors. This is often visualized through the so-called S-curve. At each stage, a different set of policy measures may be particularly effective (and will be discussed later on in this chapter).

The S-curve can be divided into three phases: innovation, growth, and maturity. A new product in the innovation phase typically experiences a slow growth rate and achieves a market share between 1% and 10%. Following this phase, the growth stage occurs, characterized by a rapid increase in market share. Eventually, the product reaches a maturity stage with a slower growth rate, typically associated with a market share exceeding 90%.

The curve may refer to the share in sales of new cars or to the full stock (fleet) registered in the country. The latter will lag over the former, but it is more relevant as for emissions, energy use and charging issues.

Figure 8

Typical diffusion curve of innovations and corresponding types of policies



In any case, it's important to analyze the introduction of a new product is to focus on the groups that express interest in the product and the market share these groups represent. The very first segment of the population to adopt the innovation is characterized as inherently innovative and is referred to as the first movers. They are distinguished by their deviation from the broader social system and by possessing a high level of technological competence.

First movers are often closely connected to the innovation process itself, and a prerequisite for becoming a first mover is access to substantial financial resources, along with the ability to absorb investments that may not yet be profitable.

An early adopter is an individual who begins using an innovation earlier than the majority within the social system. Rogers describes early adopters as opinion leaders who are respected by their peers. These individuals often serve as role models whom later adopters seek out for advice. Furthermore, Rogers characterizes early adopters as being better educated, holding higher social status, and generally being wealthier than later adopters. They are well integrated into the social system, display trust in scientific advancement, and are comfortable with risk-taking.

Within the social system, alongside the first movers and early adopters, there are also the early majority, late majority, and laggards. The early majority constitutes the largest group of adopters and tends to deliberate for some time before deciding to embrace a new technology. They represent the last group to adopt before the second half of the social system transitions. The late majority adopt technologies after the majority has already done so and is generally more skeptical towards new innovations.

The categorization of adopters follows these groups:

- The first 2.5% of the population to adopt are referred to as first movers.
- The next 13.5% are known as early adopters.
- Those within the adoption range of 16% to 50% are considered the early majority.
- Those between 50% and 84% constitute the late majority.
- The final 16% are referred to as laggards.

UNECE countries differ as for the stage of e-mobility adoption. In Norway, battery electric vehicles (BEVs) accounted for approximately 90% of all new vehicle registrations in 2024, indicating that BEVs have reached the maturity phase, where the only group still resisting adoption are the laggards.

In contrast, Sweden and Finland, with BEV market shares of approximately 30-35%, remain in the growth phase. These markets have only recently transitioned from targeting early adopters to appealing to the early majority, and significant opportunities still exist for BEVs to capture additional market share. At the same time, larger automotive markets, such as Spain and Italy, report BEV sales shares of around 5%, indicating that they have just moved beyond the first movers' stage.

For Sweden, the transition from 1% to 4% of new vehicle sales being BEVs took five years (from 2015 to 2019). Since then, the share of BEVs has increased significantly, reaching 34% in 2024, with projections indicating it will rise to 49% by 2028. If this trajectory holds, it would imply that Sweden required a total of 13 years to progress from the first movers' phase to the late majority phase.

Other countries are in further different stages. Moreover, households, companies, mobility operators and institutions participate in different shares to the total purchase and to the use of vehicles. In this chapter we focus on households, because the behaviour of households is very important for the speed and depth of the electrification of mobility.

The unsupported physiological diffusion pathway of e-mobility has several bottlenecks: a high upfront investment on the vehicles narrows the adopters by income level, concentrating it on relatively affluent households, which may prefer large and heavy vehicles, with premium features, which in turn do not suit the middle class (oriented in purchases by value for money) nor the lower income class (highly sensitive to absolute price). The insufficient large pool of EV adopters makes a large number of recharging stations not profitable, with potential industry consolidation and higher prices (to compensate the low utilization rate). All this makes e-mobility something for the environmentally conscious élites, prompting resistances from environmentally conscious middle and lower class (who may rather push for the more affordable PT) and from people not environmentally conscious. Both groups may well constitute a political majority, putting the support for public policies at risk. EVs may remain a niche, with high levels of sectoral emissions, weakening engagement by other sectors and cumulatively failing to remain within carbon budgets compatible with temperature increases that do not prompt tipping points and subsequent irreversibility.

Moreover, in many countries the entire market of new cars is already socially selective, with broad shares (or even the majority) of the purchases being of used cars (from other domestic households, from rental and business fleets, from abroad). As EVs are relatively new, in many markets there is no supply of used EV (even less of the most recent waves with increased performance and value-for-money).

In addition to social selectivity, there is typically a discrepancy between urban areas and rural regions, with the latter having often less PT available (thus being trapped between high car registration rates and immobility) and lagging behind the e-mobility diffusion curve.

It would be impossible to cover all possible conditions on this issue but here a discussion of an UNECE country can help detecting lessons, which are further developed as case study in the appendix of this report.

In short, e-mobility risks to be not accessible both for territorial and social reasons. This is not only unfair, but it produces potential resistances.

If, in contrast to an unsupported spontaneous diffusion pathway, a well-crafted strategy by public authorities and societal stakeholders, flexibly adapted to country conditions and stages of the e-mobility diffusion process, can obtain higher systemic efficiency, faster diffusion in a broader range of social and territorial groups, better matching of energy and grid opportunities and constraints, wider consensus, up to the point that electric mobility can represent a significant contribution to national and international actions (such as the Nationally Determined Contribution under art. 4 of the Paris Agreement, market and non-market mechanisms under its art. 6, etc.).

EVs have large positive (static and dynamic) externalities, by reducing GHG emissions, improving air quality and supporting the profitability of the energy transition, thus they risk having sub-optimal installation levels and rates, in absence of incentivization schemes.

Incentivization can work not only to break structural lock-in and for quantity purposes but also to shift the type of technological trajectory, for instance favouring systemic efficiency and environmental friendliness with the following co-benefits or co-features:

- Batteries without (or with low levels of) toxic and rare earth minerals;
- Vehicles capable of V2H and V2G features;
- Synergies with non-motorized transport.

At low penetration levels in the total fleet, obstacles to electrification of transport stem from the following factors:

- People do not know the EVs, they have myths and misunderstanding;
- Car dealers may push ICE vehicle sales, downplaying EVs;
- Grid operators do not consider this business as interesting;
- Lack of grid connections and EV charging opportunities;
- Certain regulations hinder introduction of light and unconventional EVs.

The current state of battery electric vehicles market in Sweden

In Sweden, new vehicles are primarily registered in larger cities and urban areas and subsequently spread throughout the country via the second-hand market. Since BEVs are still a relatively recent technology, their allocation within a country tends to be unevenly distributed, as they are concentrated in the regions where new vehicles are typically registered. Over time, however, BEVs are expected to follow a pattern like that of conventional vehicles: as they enter second-hand market, they gradually disperse to other parts of the country.

In Sweden, the national agency Transport analysis investigated if any redistribution of BEVs from urban to rural areas occurred. This was done by tracking BEVs first registered in 2020 and examining where they were registered four years later. The aim was to examine how these vehicles moved across geographic and ownership categories over a four-year period. The analysis includes both privately and corporately owned vehicles.

The geographic location of each BEV at the time of registration in 2020 and in subsequent years through the end of 2023 was tracked using the six municipal categories defined by the Swedish Agency for Economic and Regional Growth. Most vehicles were initially registered in metropolitan municipalities (Stockholm, Gothenburg, Malmö) and in dense mixed municipalities located in southern and central Sweden – 13,953 and 9,161 vehicles, respectively. Most of these vehicles remained within the same municipal category over time. By the end of 2023, 9,257 vehicles were still registered in metropolitan municipalities and 8,341 in dense mixed municipalities. While there was a notable flow of vehicles between these two urban categories, very few BEVs had moved into more rural or sparsely populated areas.

Instead of gradually diffusing into other municipal types, a significant portion of the BEVs were deregistered, particularly in 2023. Of all BEVs registered in 2020, 19% had been removed from the registry by 2023, primarily through export. The number of deregistered vehicles in 2023 alone was equivalent to the entire BEV stock in rural municipalities, underscoring the magnitude of this outflow.

The limited geographic spread of BEVs to rural areas can be partially attributed to differences in the age profile of the entire vehicle stock. The average age of passenger cars in sparsely populated municipalities was 14-17 years in 2023, making it unlikely that recently registered BEVs – typically four years old or younger – would appear in these areas. As BEVs age and change owners, broader spatial diffusion may occur in the future.

Together taken, such obstacles may lock in the system in a situation in which EVs are only adopted by very few pioneers, at the intersection of relatively high income, high pro-environmental values and penchant towards high tech solutions.

The general attitude of the population is perplexed and feel excluded (or is dismissive of possibility of considering EVs as realistic mobility solutions). In these conditions, which may be different country by country, typical venues of incentivization that might be worth exploring are the following (Type I measures):

- Information campaigns, including involving known testimonials;
- Installation of very first charging point, not only for use but also their symbolic message;
- Organization of events where all interested brands can provide driving opportunities to test the vehicles;

- Car dealers' agreements;
- Authorization of alternative, non-dealer-based, channels of distribution (e.g. online)²⁸;
- Agreements with taxi operators to highlight adoption of EVs, opening the experience of travelling in EVs for a broad swath of the population, without the need to buy;
- Platforms to share the experience of the few EV drivers, including about long trips;
- Fast-track and need-specific homologation procedure of light and unconventional, yet efficient, EVs;
- Relatively high (but time limited) subsidies for the purchase of EVs;
- Significant advantages to EV drivers, including on privileged road access (e.g. in combination with PT, as it happened in Norway).

Depending on the country conditions, the central government, local governments, the private sector and the civil society can take the lead of such activities, in the logics of cooperation and mutual interaction.

During take-off, where penetration levels in the total fleet are quickly rising, also thanks to widening model choice for value-for-money oriented buyers, obstacles to electrification of transport may stem from the following factors:

- Insufficient number of EV charging stations, possibly leading to long waiting times and even disruptions e.g. during vacation times;
- Competition in parking spaces for EVs and non EVs;
- Grid glitches in underserved areas, because of a possible mismatch between location of charging and the low voltage distribution (e.g. in residential areas with a lot of evening charging).

Together taken, such obstacles may lock in the system in a situation in which adoption is high in certain geographical and social areas but low in others, with a feeling of exclusion in the latter. Specific accidents, such as black outs, can be reframed against the energy transition.

In these conditions, which may be different country by country, typical venues of incentivization that may be explored, tested and promoted (Type II measures) are the following:

- VAT-induced price equalization or even EV getting a lower upfront price than ICE;
- Permitting simplification for new charging stations;
- Subsidy for the installation of charging station in under-served areas;
- Buy-back policies in which an objective about a certain standard in geographical distribution of EV charging stations is implemented by subsidizing the construction of EVCS where the standard is not yet achieved, and the recipient commits to pay back the subsidy after a certain number of years (during which the point become economically profitable);
- Local government policies for parking spaces, multiplying opportunities for non-motorized transport, e-bikes and e-micromobility, with clear signals that polluting cars will at a future point of time be reduced in their access to parking;
- Further measures for non-car EVs (smaller, lighter, shared, etc.);
- Reactive and pro-active grid upgrades, possibly supported by specific funding schemes;
- Specific policies directed towards creating a supply of used EVs for resale (for instance via the fast rotation of rental cars and business-purchased cars, with a platform of pre-purchasing of

²⁸ For this and several other policy approaches included in this report, see: Piana et al. (2012) and Schumann et al. (2023).

the used car at an agreed price, eliminating the uncertainty on this crucial piece of information for the first buyer);

- Low-interest loans for EV purchases or leasing programmes that lower upfront costs;
- Subsidies for low-income EV buyers; they will also be favoured by the availability of low-cost brands and models, with smaller batteries, requiring a denser system of locations for charging;
- Mandatory rural charging deployment;
- Zoning reforms for integrated infrastructure.

Depending on the country conditions, the take-off may occur in synchrony with neighbouring countries or with leakage effects (a large foreign pool of EVs is an important customer for local EV charging stations, whose diffusion engender a similar imitation effect in the country). In this case, some cooperation across neighbourhoods may be particularly effective.

At high penetration levels in the total fleet, especially if in general there is a high motorization rate (in which a relatively large share is in big vehicles), electrification of transport may be characterized by the following inefficient situations:

- Frequent congestion in many parts of the grid;
- Long queues at public charging stations;
- High demand during certain hours, putting a stress if those hours are the same as the peak for other types of electricity consumption.

Together taken, the system may be in a situation in which laggards feel vindicated in their resistance to change and are more committed to remain at ICE vehicles. However, they are a minority and may need to be pushed towards abandoning polluting habits, especially if their reasonable objections are taken into account. Bans would be effective at this stage and not too unpopular.

In these conditions, which may be different country by country, typical venues of incentivization (Type III measures) are the following:

- Sufficient expansion of the grid, including smart local grids;
- Wider range of locations where to charge;
- Vertical and horizontal road signs to charging point to redistribute charging opportunities;
- Hourly-differentiated prices to reduce the peaks;
- An exit-strategy from gas stations;
- Removal of obstacles and specific incentives to couple PV, storage and charging at home;
- Low emissions zones, possibly surrounding pedestrian areas, providing relief to air quality issues.

Depending on the country conditions, these policies can be enacted in anticipation of high penetration, already in the take-off stages.

In summary, across the three types of measures, we considered different families of incentivization measures and strategies:

- monetary incentives to purchase a vehicle or install infrastructure;
- liberalization measures for allowing pioneer households and manufacturers to be able to act;
- soft informational policies and nudges;
- policies that facilitate experience gaining and sharing, operating shifts in perception and self-assessment;

- selective technological interventions by operators and stakeholders (including investments, which will be further developed in another section of this study);
- Mandates that oblige or ban certain situations or behaviours.

They have the following pros and cons:

- monetary incentives are a part of public expenditure, with the corresponding costs for taxpayers and procedures; they are quite effective in providing hard facts in favour of a solutions, while sometimes being quantitatively too small to overcome certain groups resistances, in other cases being windfall profit for people that would have moved anyway, in still other cases they are so generous that a point they need to be retracted;
- liberalization measures may require administrative and regulatory innovations and being not allowed for different sets of reasons; while being effective for committed pioneers they may sometimes not being enough to convince the early majority or later groups;
- soft informational policies may be insufficient to move behaviour and not fully understood and undertaken at a high-quality level, while being relatively cheap;
- policies targeting experience require fine-tuning and cooperation across different scales, so that the experience is relatively consistent;
- selective technological interventions are in line with certain industrial policies but may raise resistance in environments that are systematically technology neutral;
- mandates require enforcement capacity; they are effective and cheap but often face political resistance.

It is very likely that no single measure is a silver bullet that automatically assures an expansion of e-mobility. It is rather a package of different measures, some announced for future dates, others widely expected and implemented with sufficient gravity, that will significantly accelerate and broaden the appeal of e-mobility.

Governments that prefer not to spend money may explore in detail in particular regulation mandating certain behaviours, informational and experience-targeting policies, coordination with stakeholders through moral suasion, so to mobilize their resources.

Governments that accept to make investments with taxation may explore in detail in particular subsidies and buy-back policies (the latter being fiscally neutral over a certain amount of time).

Government in countries without local manufacturers, thus importing all vehicles, can utilize import regulations and voluntary agreements.

Countries where one (or few) large cities concentrate the bulk of population and employment may consider empowering such cities with regulatory powers and implement very local policies (e.g. parking allocations) having national implications. Conversely, the mayors of such cities may consider an alliance with the same effects.

Countries with large swaths of rural areas can target farmers and small villages to support widespread adoption of chargers open to the public but predominantly used by local people.

Countries with significant inflows of international tourist may have a strategy of dual use installations (with most cost covered by tourist and free/cheap charging for locals). Liberalization and branding orchestration may allow inhabitants that do not have EVs to offer charging from home for a profitable price, e.g. using a nationally designed vertical sign indicating this (income-generating and useful to sustainable tourism) opportunities to passers-by. More generally, actual designs of policies should reflect national specificities, fitting multiple goals and strategies.

Changing mobility patterns and modes through location efficiency

The location of people across residency, workplaces, shopping opportunity depends on urban forms and spatial organizations. The longer the distance between destinations and origins, the more unpredictable the timing of trips, the lower the number of people involved, the higher the probability that the trip will be motorized and, in many situations, carried out by cars. This boosts VKT as well as personal kilometres travelled. So systemic efficiency can be related to the location of people, before and beyond the location of EV chargers, as it was discussed previously.

Policies that would promote such interpretation of location efficiency are the following:

1. Encouraging mixed-use development
 - a. Promote urban planning strategies that integrate residential, commercial, and recreational spaces within walkable distances.
 - b. Reducing travel distances that encourages active mobility (walking, cycling) and reduces car dependency, including by providing comfortable benches and opportunities to rest and be protected from challenging weather conditions.
2. Enhance public transit accessibility
 - a. Improve PT routes, frequency, and reliability to make it a convenient and attractive alternative to private vehicles.
 - b. Introduce transit-oriented development (TOD) principles, widened by ensuring easy access to transit stations through EV charging and pedestrian-friendly infrastructure.
3. Promote shared and MM solutions
 - a. Support bike-sharing, e-scooters, and e-carpooling systems in urban centres to provide flexible, last-mile connectivity.
 - b. Remove obstacles to shared and MM solutions, in terms of parking spaces, safety, competitiveness with cars.
 - c. Implement smart mobility hubs where users can seamlessly switch between different transport modes.
4. Leverage digital tools and behavioural nudges
 - a. Utilize mobile apps and real-time data to provide commuters with personalized, efficient route options and sustainable transport incentives.
 - b. Introduce policies like congestion pricing, parking restrictions, and incentives for off-peak travel to shift mobility behaviours;
 - c. communicate how important for range is to keep the window closed and to have EV-specific tires.

In parallel, adding bike stalls and EV charging where people would have gone in any case, without the need of long detours, is a bottom – up measure conducive to more efficient mobility patterns.

Peak hour considerations and temporal aspects of charging

People are very sensitive to traffic disruption and delays, as main perceived direct annoyance of mobility. This has fostered, in many countries, large investments in roads, offering temporary relief but encouraging further the dependence of cars, especially if designed without complete streets and the needs of vulnerable users in mind and excluding bikes and MM. This also tend to attract competences, finance, and political attention. Sustainable mobility, which often requires a shift away from cars, needs to consider both the short-term and the long-term requirements of people, including by reducing the time to transit via alternative means.

By reducing congestion, the quality of life of people (even measured subjectively with happiness questions) is improved and the pressure for further developments of car-centric society and spatial planning is reduced.

In this vein, the following groups of policy measures might be evaluated:

- a. Encourage Flexible Work and School Schedules
 - a. Promote staggered work hours, remote work, and flexible school start times to reduce peak-hour congestion, adopting flexible hours, reducing rush-hour traffic, staggered school start times (8:00 AM to 10:00 AM), longer hours for shopping centres can decrease morning or evening peak congestion.
 - b. Employers and institutions can implement rotational shifts to distribute travel demand more evenly all day.
- b. Incentivize Off-Peak Travel and EV Charging
 - a. Offer lower fares for PT during off-peak hours to encourage commuters to shift on travel times.
 - b. Implement congestion pricing strategies to discourage peak-hour private vehicle use and promote alternative travel times.
 - c. Congestion tax on area basis can reduce peak traffic. Commuters can also be offered "Happy Hour" metro discounts (for example 25% off 10:00 AM–3:00 PM).
- c. Enhance Real-Time Traffic and Transit Information
 - a. Use digital platforms to provide commuters with real-time traffic data and alternative route suggestions, including EV charging opportunities.
 - b. Mobile apps & dynamic signage can help redistribute traffic flows by directing users to less congested options.
- d. Promote Active and Shared Mobility During Peak Hours
 - a. Improve infrastructure for walking, cycling, and MM (e.g., bike lanes, pedestrian pathways, parking spaces)²⁹ to reduce reliance on motorized transport.
 - b. Expand shared mobility options, such as e-carpooling, ridesharing, and park-and-ride facilities, to decrease single-occupancy vehicle trips during rush hours. E-scooter partnerships with discounted "peak avoidance" rides can improve traffic flow during peak hours.

All this should foster a smoothing of the peaks, with benefits to the participating people, who will enjoy shorter trips. However, their total emissions will not fall unless a modal shift and/or a shift towards EVs happens. In this latter situation, a minor, yet not irrelevant, issue is when charging happens, since the carbon intensity of the electricity mix depends not only on the country but also on the hour.³⁰

When targeting shifts in timing of activities, a particularly relevant issue is related to the behavioural rules for EV charging (timing and location).

In this vein, it should be underlined that the list of locations for charging indicated at the "Location efficiency and its various contributing factors" greatly differ as for the timing in which they tend to be used. Households are at home, on the road, at employers' premises, commercial premises, leisure

²⁹ See: <https://parkedbyme.com/>

³⁰ See: <https://app.electricitymaps.com/>

facilities, at their different destinations during the different hours of the day, also depending on seasonality.³¹

People typically recharge after using the vehicles, but timers, apps and smart devices could postpone and manage the actual timing and speed. Smart charging has been proposed as a key solution to this problem. Here it is important to underline that smart charging can be achieved with social rules, in addition and complement to technology. A crucial information for smart charging is to know “when the next trip will start” and “how long it will be” (until next occasion to charge), so that any postponement or speed slowing down can be compatible with users’ need. AI and training can provide a reasonable guess but some explicit agreement with the user will probably be needed for actual acceptance of the system. It can be anticipated that people may be willing to accept smart charging if it does not interfere with their plans, their requested flexibility to face uncertainties, and provide some tangible benefit.

From a territorial perspective, if a certain grid is congested in certain peak hours, locations for charging that tend to generate load off-peak should be preferred.

A chrono-infrastructure approach, determining when a certain system of locations will cumulatively solicit the grid, needs to take into consideration that people are learning to use their EVs so proposed social rules, campaigns and incentives can still shape this crucial factor.

Cultural and regional influences on mobility choices

A mobility model based on “new cars” is the mainstream situation in many countries. There a larger share of e-cars on sales is the main venue for electrification of transport overall.

However, in other countries (and for certain segments of populations) there may be the dominance of old cars, two-wheelers, PT, and paratransit. Accordingly, the electrification of mobility can occur in such other types of vehicles, sometimes even faster than with reference to standard cars. With the exception of old cars (for which some would consider technical conversion of ICE vehicles to electric), all other modes are more efficient than cars (although in certain conditions less comfortable), so if electrification deployment may help switching to such modes, away from cars, we would have a double dividend in system efficiency.

In addition to objective facts about ownership, modal choice is strongly influenced by culture, with some groups having developed a sort of “worshipping” of cars as symbols of freedom and economic success, whereas others tolerate some disadvantages of PT in view of cherishing personal contacts and commonalities with other people. Mono-modal people have cognitive and emotional structures that devalue what is not chosen, to the point of ignoring a lot of details that would be necessary to use it, with a strong confirmation bias and real impediments. For instances, monomodal car-centric people may completely ignore PT timetables and underestimate the time passed in cars (attributing to traffic jam large part of it). Conversely, a growing number of people is becoming multi-modal, supported by apps that offer multimodal trips (e.g. sharing a bike until a certain PT hub).

In Europe and North America, people own in majority private cars, although they often can choose PT (whose electrification is at the order of the day) and cycling (in rural areas and strongly promoted in

³¹ See: <https://www.bls.gov/tus/data.htm>; animated data visualization of time series is also available at: https://www.linkedin.com/posts/jameseagle_its-400-am-while-linkedin-warriors-post-activity-7259074415919853568-Lo8u?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAbsz4BfeljJASv4t2FORoIRm2kNRAkY1Y

many cities). Here modal choice is the key, with nuanced incentives and disincentives on cars in general, in certain cases irrespective whether they are electric or not, in other cases with privileges for EVs. For instance, night transit on certain roads may be forbidden for noise pollution reasons. At certain speed, EV are significantly less noisy, so they might be allowed to pass. In other cases, only bikes and PT is to be allowed.

In South and Southeast Asia, cars are less common, and two-wheelers (motorcycles, scooters) are the main private mobility means, supplemented by PT. Electric scooters, including with swapping batteries, can represent a major improvement for air quality, actual mobility, and transport of goods. By contrast, a policy centered on electric cars would not avoid congestion, conflict for space and particularly serious accidents.

In some parts of Africa, paratransit (minibuses, shared taxis), formal and informal privately-owned collective vehicles (with or without timetables) fill the gaps left by limited PT, bikes and walking for long distances. Here, a redesign of such collective vehicles, including electrification but also adapting for climate, mixed good and personal transportation, and of unpaved roads may be conducing to a system of mobility that may not necessarily be car centric. The electrification of two-wheelers and cargo bikes is consistent with such potential perspective of original sustainable mobility.

In the Balkans and in the Middle East, but not only, there are areas highly congested with old cars, often imported. Any ban would be extremely unpopular but measures for a transition (including by developing a market of second-hand EVs) are probably called for.

Policies for creating a second-hand car market (domestically and internationally) are actually of interest for many countries. The policy might target taxi fleets, company fleets and rental car fleets for purchases of new car under binding commitments for reselling after two or three years of use at an adjusted price to an agreed household. In this way, the businesses costs are certainly only of the difference between the two prices and people have two or three years to cumulate the savings necessary to the used car purchase.

More in general, it's important to underline that, in contrast to certain technological discussions whereby car centrality is taken for granted, sustainable mobility, including electric mobility, should be ready to embrace the variety of cultural and regional differences in order to evolve the current systems towards directions that are not necessarily car centric.

Shared mobility to reduce private mobility needs

In most of this report, the focus has been on private cars owned by households. However, in the last few decades, an alternative to private ownership has been proposed by operators that have a fleet and allow rental by the hour (or even counting the minutes). Since the normal use of a car is for a small part of the day and night, sharing companies can rent to several people the same car in the same day, potentially arriving to a price that is lower than the private car use - especially if the vehicle is electric (for its low "fuel" cost), the traffic flows are balanced (without too much need of redeployment), vandalism is kept at bay and public authorities provide conducive conditions.

Not always such conditions hold, with the consequence that sharing may turn out to be significantly more expensive than the use of private car, so displacing it only in premium locations and occasions (e.g., to come from work after having reached it by PT, with the desire of flexibility for the evening).

Local and regional authorities and stakeholders, including large employers, can play a role in evaluating and shaping sharing systems so that they reach their most ambitious goals of substituting a large share of trips (or even the need for ownership).

In this vein, policies that might be evaluate include the following ones:

1. Expand and Integrate Shared Mobility Services
 - a. Promote EV adoption in car-sharing, ride-hailing, and bike-sharing services to reduce reliance on private car ownership.
 - b. Ensure seamless integration between shared e-mobility and PT, allowing for easy multimodal commuting.
2. Encourage Workplace and Community Ridesharing
 - a. Support corporate and community-based e-carpooling programmes to reduce single-occupancy vehicle trips.
 - b. Use digital platforms and incentives (e.g., priority lanes, discounted tolls, symbolic reciprocation) to encourage shared commuting.
3. Leverage E-MaaS for Efficiency
 - a. Foster MaaS platforms that combine multiple clean transport modes into a single, user-friendly app for planning and payment, including with opt-out logics.³²
 - b. Offer subscription-based shared e-mobility packages to reduce individual travel needs and improve cost efficiency.

Addressing perceived risks of e-mobility

Its large swath of advantages notwithstanding, electric mobility has an history of attracting several criticisms. Some of them have been addressed (e.g. with improved technologies), others remain in the public and expert debate, requiring fine-tuning of the proposed advancements.

Public skepticism about EV battery safety persists despite rigorous European Union regulations. For instance, the European Union Battery Directive (2023) mandates stringent crash-test certifications and thermal runaway prevention systems. However, the public perception may be influenced by the fact that wide coverage is given to EV or battery fire accidents. Campaigns with comparative data may be helpful to enhance public confidence and adequate behaviours by drivers and fire brigades.

It will be important to provide clear and accessible information on the safety, reliability, and benefits of e-mobility, including battery safety and environmental impact. The latter has a lot to do with the specific working and production conditions of certain countries, where pollution is somehow *de facto* tolerated. Improvements in labour conditions and position may complement other developments (e.g. reduction or elimination of problematic raw materials or sources).³³ It is worth exploring the mandate to keep compatible replacement batteries for a long number of years and to provide transparent information on battery lifespan and replacement options.

Potential venues to address perceived risks of e-mobility are the following:

1. Strengthen Charging Infrastructure and Range Confidence
 - a. Expand the availability of fast-charging stations in a broad range of locations to address range anxiety.
 - b. Implement real-time charging station locators and reservation systems to improve accessibility and ease of use.

³² For a multi-country project enhancing MaaS, see: <https://www.interregeurope.eu/maasolutions>.

³³ See: <https://projectcobra.eu/> (European Union project aiming at developing the next generation of Cobalt-free Li-Ion batteries for electric vehicles).

- c. Fast-Charging Networks and Smart Data: It can drastically reduce charging time. Technology is fast changing trying to bring fast charging within few minutes. This will improve public confidence.
- 2. Address Misconceptions on Cost, Maintenance, Comfort, Conflict with Public Transport
 - a. Highlight long-term cost savings of EVs, including lower fuel and maintenance costs, through comparative studies and experience sharing platforms;
 - b. conduct public demonstrations and test-drive events to increase familiarity and personal experience;
 - c. underline the alliance across PT, trains, e-bikes and MM in what is promoted for the sake of noise, health and environment and affordability.
- 3. Grid Resilience Through Renewables and V2G
 - a. Support synergies with PV, so assuage the perceived risks of grid congestion
 - b. Solar-powered charging stations can offset grid demand.
 - c. V2G and bidirectional chargers to feed surplus EV electricity back during peaks, providing participants with monetary and non-monetary incentives.
 - d. data sharing for energy communities, virtual power plants and other actors to step up actions in critical moments;
 - e. international funds for cross-country grid-PV integration;
- 4. Rural and Cross-Border Solutions
 - a. Reliable rural and cross border EV charging infrastructure along with standardization of cross-border payment systems;
 - b. develop local and cross-border smart grids to facilitate charging.

INVESTMENT

Business models to support e-mobility adoption

This chapter addresses concisely the role of private sector and the broader institutional environment in which electric mobility is taking off. In particular, it addresses the key issue of “who pays for what” and the related externalities that are covered by somebody else (or lead to deceleration and acceleration of the dynamics). The main private actors are the manufacturers of vehicles, backed by financial markets (with their erratic behaviours), typically remunerated by the sales to households. However, in some countries, many obstacles to households’ purchase (including low awareness and attributed value to environmental benefits, income limitations, settlement limitations as for house typology and connecting areas, social prejudice) have been reducing the size of the market, with the effect of limited profitability of the manufacturers. Foreign markets may be more promising but tensions on tariffs risk to fragment the world market and destabilize companies’ plans.

All this increases the possible role of governments, which, in the logics of green public procurement, support to infant industry with high environmental positive externalities, fulfilment of Paris Agreement and SDGs objectives, can become early adopters of vehicles. Further important private sector demand sides are companies (for their fleet) and logistic operators, possibly convinced by requests from their downstream customers (such as supermarkets). Sharing companies can adopt one or more brands and models in fleets that the general public uses “in small doses” for a very low entry price, allowing for gaining experience. Banks and leasing institutions can alleviate the funding conditions for the purchase and back the commercial proposition of the sellers. A second role for the government, in addition to direct purchase, can also be to orchestrate and facilitate such developments. This is particularly important for non-car EV, such as e-micromobility, e-bikes and others, because of their high systemic efficiency but the wide range of obstacles they meet. The cooperation with the PT, which often relies on subsidies, is very important and need to be conducive to higher occupancy rates.

Direct support to manufacturers in establishing R&D and production facilities (for vehicles and their key components, such as batteries) is becoming common; to level the playing field is to be evaluated that all countries engage, according to their present or desired role in global value chains, in such supporting activities. Supporting the households in their purchase (rental or lease) is often a tempting venue to spread e-mobility.

In case of business failures, it should be evaluated whether the company developed valuable technologies and plans, organizing a future for them, maybe in the hands of other companies or investors, in order for the regions hosting the company to continue to play a role in the international developments.

Supporting the conditions for a wide market is at the end the most comprehensive means to assure that the spontaneous dynamics for profitability of the manufacturers deliver, by economies of scale and scope, lower prices and wider choice of models.

Vehicle manufacturers are not the only private actor key to e-mobility. CPOs are companies that develop EV charging stations and charging support equipment, install hardware from a wide range of EVSE vendors, and ensure optimal ongoing EV charging operations. They bet on fast deployment of vehicles and on having accessed remunerative locations. They are hindered by long and uncertain permitting processes, expensive connections to the grid, unfavourable logistic circumstances. If the number of EV does not grow as forecast or the share of public charging is too low, interest rates on debt rise, CPOs may go bankrupt, with consolidation trends across the industry. While seamless

roaming and the multiplication of locations, especially in places where there is a fairly good connection to the grid and to amenities, are contributors to the take-off, excessive concentration of the industry may lead to high prices, cutting the savings that buyers expect to compensate the higher upfront cost of the vehicle.

Accordingly, business models of cost and revenue sharing with the cities and amenities can bridge the industry towards the turning point of such large EV adoption at which shopping areas see a business advantage in offering availability and reasonably priced recharge. At that point everyone will have the certainty of reasonably ubiquitous charging opportunities (in addition to home charging, treated in another part of this document).

In other terms, the three questions (1. Who pays for the vehicle? 2. Who pays for the charge? 3. Who pays for the charging infrastructure?) can have different answers. The conventional answer is that the vehicle is owned and paid by a household, the charge by the owner of the vehicle, and the infrastructure by profit-oriented private specialized operators. Unconventional business models cover for instance cases in which charge is for free, paid by parking operators (thus bundling with the parking space fee), or paid by restaurants by which the charge is taking place (thus implicitly bundling with meal expenditure), or paid by supermarkets. In all these unconventional cases, e-mobility becomes cheaper, charging more accessible, thus more widely adopted. This is even stronger if it is the vehicle that is paid not by the user but by other entities (as it happens with company cars).

Energy communities of citizens and companies can integrate EV charging in what is shared and exchanged (for a price or other form of remuneration), including in logics of club extending renewable energy supply and out-of-the-road parking to members and, under certain conditions, non-members. Managing EVs collectively (e.g., in intelligent parking lots) allows communities to exploit aggregated storage and flexibility, optimizing both energy use and economic returns. Energy communities can effectively optimize electric mobility integration through advanced socio-technical management, smart charging, and coordinated infrastructure planning.

Across these venues, and others not mentioned for brevity's sake, significant investment in systemically efficient locations is to be expected.

Long-term returns on sustainable mobility investments

The global mobility landscape is undergoing a profound transformation, driven by the convergence of EV technology, renewable energy, and digital infrastructure. These advancements present a significant opportunity to transition to sustainable, location-efficient mobility systems. While the upfront costs of such systems may be substantial, the long-term financial, environmental and health-related returns make a compelling case for investment.

A growing market opportunity

The sustainable mobility market is projected to reach USD 1.5 trillion by 2050, fueled by innovations in electric mobility, battery storage, and smart energy systems (IEA, 2022). Global EV sales are growing at a rate of 55% year-on-year, reflecting increasing consumer demand, favourable policy frameworks and technological maturity (IEA, 2023). However, to sustain this momentum, the supporting infrastructure—particularly EV charging ports—must expand significantly. For example, the United States alone is projected to require 12.9 million public and private charging ports by 2030, underscoring the scale of infrastructure investment needed worldwide (United States Department of Energy, 2022).

Beyond EVs, emerging technologies such as Electric Road Systems, smart grid integration, and V2G solutions offer substantial efficiency gains. These innovations enable real-time energy demand

management and allow EVs to act as decentralized storage units, supporting energy resilience and grid stability (IRENA, 2019).

Financial and economic returns

Investments in sustainable mobility consistently outperform traditional fossil fuel assets. Renewable energy investments – including those that power electric transport systems – yield an estimated 16% compound annual return, higher than many conventional energy portfolios (Bloomberg NEF, 2021). As wind and solar power become increasingly cost-competitive, integrating these sources into mobility systems enhances financial viability.

In terms of productivity, sustainable mobility systems generate substantial economic returns through improved public health. Reduced reliance on combustion engines leads to cleaner air and fewer road accidents, while the promotion of active mobility contributes to healthier populations. Estimates suggest that investment in active transport infrastructure could save USD 1,170 per capita in health-related costs by 2050 (WHO, 2017).

The value of reduced fatigue and stress from improved transit systems—such as shorter commutes, reliable PT, and less traffic congestion—can contribute billions to national productivity. In the United States, reduced sleep deprivation and travel-related stress from sustainable mobility investments are projected to deliver billions in economic benefits annually.

Environmental and climate impacts

From a climate perspective, electrifying even a small proportion of road infrastructure yields exponential benefits. Research indicates that electrifying just 3% of road networks could reduce global transport-related carbon emissions by up to 60% (UNEP, 2021). These emissions savings are critical to meeting the goals of the Paris Agreement and achieving long-term energy transition targets.

Sustainable mobility also complements land use and location efficiency policies, promoting compact development and reducing urban sprawl. This in turn minimizes the need for road expansion and preserves green spaces, contributing to broader environmental sustainability.

Building future-proof mobility systems

The long-term success of sustainable mobility investments depends on future-proofing strategies. Key pillars include:

- Digital optimization: AI and data analytics can enhance route planning, energy management and user experience.
- Public-private partnerships: Collaboration between governments, industry and civil society ensures shared investment and operational responsibility.
- Robust policy frameworks: Clear, stable and long-term regulations encourage investment, innovation and equitable service delivery.

Governments should prioritize equity and accessibility in mobility planning. Ensuring that low-income, rural and marginalized communities benefit from clean, efficient transport options is essential to delivering just and inclusive development.

Conclusion

Investments in sustainable mobility represent not only an environmental imperative but also a sound economic and social strategy. From reduced carbon emissions and improved public health to resilient infrastructure and job creation, the long-term returns significantly outweigh the initial costs. Policymakers are encouraged to align transport, energy and urban planning to realize these returns,

ensuring that both current and future generations benefit from cleaner, safer and more accessible mobility systems.

EV charging carbon credits for sustainable mobility

The convergence of sustainable mobility, clean energy and climate finance presents new opportunities to accelerate the transition to net-zero transport systems. At the heart of this transformation is the rise of EV charging infrastructure, which is not only enabling zero-emission transport, but also catalyzing access to carbon credit markets. By monetizing the emission reductions associated with EV charging, carbon credits offer powerful financial incentives for clean transport deployment.

Growing demand and infrastructure gaps

Public interest in electric mobility is growing rapidly. As of 2023, 62% of adults globally expressed interest in purchasing EVs, a trend driven by environmental awareness, policy incentives and advancements in EV technology. However, infrastructure remains unevenly distributed. While urban centres have seen accelerated deployment of charging networks, rural and underserved communities continue to face access barriers, limiting inclusive adoption.

At the same time, technical innovations are making charging faster and more efficient. Ultra-fast charging stations are now capable of delivering 200+ miles of range in under 15 minutes, making EVs more viable for a wider range of users and use cases.

Carbon credits from EV charging: framework and process

Carbon credits linked to EV charging infrastructure are generated when electricity displaces fossil fuel use in transport. Each kilowatt-hour delivered to an EV avoids emissions that would otherwise be produced by internal combustion engine (ICE) vehicles. The process of generating carbon credits involves four primary steps:

- **Generation:** As EVs charge from the grid, they reduce tailpipe emissions. These reductions are converted into credits based on the amount of carbon dioxide (CO₂) avoided.
- **Calculation:** Emissions reductions are calculated using several variables, including the carbon intensity of the local electricity grid, the efficiency of the EV, and the baseline emissions of ICE vehicles.
- **Verification:** Third-party auditors validate the emission reductions. Standards such as the Verified Carbon Standard (VCS) ensure environmental integrity and prevent double-counting.
- **Trading:** Once verified, credits can be sold on voluntary or compliance carbon markets. Depending on location and usage, each EV charger can generate USD 249 to USD 998 in carbon credit revenue annually.

Incentives and deployment benefits

Carbon credits provide tangible financial incentives for EV charging infrastructure operators. Beyond user fees, station owners can tap into recurring income streams, improving the business case for investment. In regions where infrastructure roll-out has been economically challenging, this additional revenue can accelerate deployment and enhance returns on investment (ROI).

For example, multi-unit residential buildings (MURBs) in Canada have reported annual revenues of USD 8,000 from just four Level 2 chargers, while condo properties in British Columbia have achieved a 400% ROI improvement by participating in carbon credit programmes. Additionally, the Connecticut Green Bank has successfully aggregated credit generation across more than 100 local partners, creating scale for smaller operators and enhancing regional impact.

Enabling grid integration and smart charging

Carbon credit schemes also reward smart charging practices that align EV demand with periods of high renewable energy generation. This improves grid efficiency and flexibility, while further reducing the carbon intensity of charging. Moreover, V2G technologies offer emerging opportunities to treat EVs as distributed energy resources. By exporting stored power back to the grid during peak demand, EVs can generate additional credits, deepening their integration into clean energy ecosystems.

Outlook and market opportunities

As countries adopt net-zero transport strategies, EV charging carbon credits are expected to play an increasingly central role in climate finance portfolios. Forecasts indicate that carbon prices could rise substantially by 2030, enhancing the economic viability of credit-generating activities. Policymakers and stakeholders can be encouraged to integrate carbon credit mechanisms into transport and energy plans, particularly in areas with infrastructure gaps. Doing so not only supports emission reduction goals but also ensures more equitable access to clean mobility solutions.

LAND USE

Mixed-use zoning

Mono-functional zoning, separating residency from employment, education, recreation and high-ranking mobility hubs, can be suitable for developing differentiated urban standards and conditions for life. However, it exponentially increases intra-city mobility, not always manageable by PT. Carefully assessed to avoid incompatible uses, mixed-use zoning (with vibrant life without too much noise, employment opportunities without pollution, commercial services in proximity of residency) drastically reduce the need for short and mid-range mobility. By dynamically operating on existing neighbourhoods with insertions and deletions, incentivized by offering opportunities inside and outside the area, mobility can become more sustainable, with larger share of walking and cycling.

The substitution of fossil fuel station with EV charging stations reduces soil pollution, allows for denser housing development nearby, and increase the opportunity of partial charging, which is faster (and better for battery durability).

Urban sprawl vs. compact cities: understanding the impact of land use patterns on transportation demand and emissions

The expansion of cities at their peripheries, with wider and wider range of required mobility to reach city centres and employment opportunities has been at the centre of policy attention since several decades. The push towards “urban sprawl” is a combination of intolerably high prices at the city centre, squeezing families in smaller flats, often high pollution and lack of green spaces, difficult life for children in their walks across congested and dangerous roads and of speculative transformation of agricultural land in residencies. As it may turn out that PT from such enlarged peripheries is not viable, the predominant mode becomes the car, whereby people living more and more distantly from the city enter and congest a city, whose inhabitants may be in turn relying on PT and renounce to the car, also because of the high costs of parking.

A more compact city would require a lower VKT and lower emissions. Higher density, mixed land use, and better street connectivity promote walking, cycling, and public transit use, lead to shorter commuting pathways and time and more sustainable mobility patterns. For it to happen, planning restrictions and regulations are necessary, but not sufficient. An answer should be given to the high cost of housing; small but ubiquitous green spaces, pedestrian and low-speed (or no-emission) zones should be guaranteed, also with the rebalancing of street space allocation.³⁴ In this context, which goes beyond mobility, e-mobility can play a very positive role. By reducing noise and air quality pollution, in alliance and not contraposition with bikes, pedestrian facilitations and PT, e-mobility can make cities more livable. Mobility hubs and exchange parking with PT and shared e-micromobility at the outskirts of the city can deviate traffic and prevent congestion.³⁵ They could also serve as hub for EV recharging. More innovatively, small electric buses, maybe autonomous, might reconstruct the conditions for zero-emission PT to be economically viable and offered in all the peripheries. Widespread territorial

³⁴ On the comparative experience of cities in such rebalances, see: <https://reallocatemobility.eu/>.

³⁵ For a transformative logistics solution for local commerce based on a collaborative approach between cities, local commerce, logistics operators, e-commerce platforms and e-vehicles sharing platforms see: <https://tacticlogistics.eu/> (European Union project TACTIC). The main objective of this project is to establish a sustainable and efficient last-mile delivery system at the city level. To achieve this, two Living labs are designed and implemented in Barcelona and Metropoli du Grand Paris.

distribution in the periphery of services, maybe in a reduced and simplified manner such as the durable good installed with the DUT project Multigeneration, can reduce the need for people to move to long distances from their homes.

In synthesis, urban sprawl is not systemically efficient as it generally increases travel distances, emissions, costs, and reliance on private vehicles, while reducing access to opportunities. Compact, well-connected urban forms, with equitable solutions to city issues, support more sustainable and efficient mobility.

Rural land use considerations: addressing the unique challenges of mobility and land use in sparsely populated areas

Introduction: rethinking rural mobility

Sparsely populated regions, particularly in rural areas across North America and globally, face unique land use and mobility challenges. In contrast to dense urban environments, rural areas must confront the spatial and demographic realities of longer travel distances, lower population densities, and limited transport infrastructure. These challenges are compounded by ageing populations, socio-economic disparities, and declining access to public services.

More than one million rural households in the United States, for example, lack access to a personal vehicle, while PT availability remains critically low – only 60% of rural counties have any PT service (United States Department of Transportation, 2020). As trip distances in rural areas have increased by 12% since 2001, mobility has become an acute barrier to inclusion, especially for vulnerable groups such as the elderly, children, low-income families, and persons with disabilities.

Addressing rural land use and mobility requires an integrated, context-specific approach. It must balance modern technological solutions with rural character preservation, while ensuring that location-efficient development principles are adapted to low-density settings.

The rural mobility landscape: barriers and opportunities

In rural regions, access to essential services, such as healthcare, education, and food retail – often requires trips of over 30 km (20 miles). In the absence of adequate public transit, car dependency becomes the default, yet not universally attainable. This exacerbates inequality and creates a mobility gap that deepens rural isolation.

Infrastructure in rural areas is typically car-centric, with minimal provision for alternative modes of transport. Roads often lack sidewalks, shoulders, and bicycle lanes, increasing risks for non-drivers. At the same time, rural populations have 20% more elderly residents than urban counterparts, underscoring the urgency of inclusive and safe mobility options (UN-Habitat, 2020).

Land use patterns also hinder accessibility. Dispersed development, characterized by scattered housing and isolated amenities, increases the cost and frequency of travel, while single-use zoning prevents the creation of walkable, mixed-use town centres. In many cases, rural areas suffer from land opportunity costs, where excessive road width and parking areas consume valuable land that could otherwise support compact development.

Towards sustainable land use and mobility integration

To foster location efficiency in rural areas, land use planning must prioritize compact and clustered development. By concentrating essential services – health clinics, markets, administrative centres – within rural town cores, trip frequency and length can be reduced. This supports both economic

resilience and social cohesion by making daily needs accessible without reliance on long-distance travel.

Integrated land use and transport planning is vital. New rural developments should be subject to mobility impact assessments, ensuring that housing projects are supported by viable access routes, shared transport services, or proximity to community services. Where possible, brownfield redevelopment and infill strategies should be prioritized over greenfield expansion to preserve agricultural land and reduce infrastructure costs.

The role of EVs in rural settings

The global shift toward electric mobility presents both challenges and opportunities for rural communities. While EV adoption in rural areas remains low due to range anxiety, high upfront costs, and sparse charging infrastructure, these challenges can be overcome through strategic interventions. Importantly, rural areas possess abundant solar and wind resources, positioning them to become energy-resilient microgrids capable of supporting EV infrastructure (IRENA, 2020).

Installing charging stations at community hubs, such as town halls, post offices, and grocery stores, can provide essential coverage and encourage uptake. Additionally, the integration of EVs with local renewable generation supports energy independence and economic diversification. Nonetheless, rural power infrastructure often requires significant upgrades to support high-capacity charging. Investment in grid modernization is therefore essential, supported by coordinated planning between utilities, municipalities, and rural development agencies.

Innovative solutions: adapting technology to local needs

A growing number of pilot programmes demonstrate how digital innovation can enhance rural mobility. On-demand transit, facilitated through mobile applications or community call centres, provides responsive transport services tailored to fluctuating rural needs.³⁶ Shared mobility models, including community-owned vehicle fleets, reduce the cost burden of car ownership and maintenance. Investments in active transport infrastructure, such as walking paths and cycle lanes within rural town centres, promote healthier, low-cost travel. Teleservices, including telemedicine, distance learning, and e-governance, can also reduce the need for physical travel altogether, especially in remote or underserved regions.

These approaches must be supported by dedicated rural mobility funds and grant programmes to pilot scalable, community-driven solutions. Long-term success depends on maintaining flexibility, encouraging innovation, and respecting the unique identity of rural communities.

Policy and planning recommendations

For rural areas to thrive in the context of sustainable mobility and land use, targeted policy measures to evaluate can be the following:

- Integrated Planning: Align transport infrastructure with land use zoning, requiring coordination across housing, energy, and mobility sectors.
- Energy Communities: Support community-owned renewable energy systems and microgrid development, especially in off-grid areas.
- Development Incentives: Offer tax credits or subsidies for infill and mixed-use projects that bring services closer to residents.

³⁶ See, for example: <https://www.nemi.mobi/flexible-rural-transport-solutions/>

- Funding Mechanisms: Establish national and subnational funding windows for rural mobility innovations, infrastructure upgrades, and EV adoption.

Above all, a community-led approach is vital. Residents must be involved in the planning, design and operation of mobility solutions to ensure local ownership and sustainability.

Conclusion: a holistic path forward

Rural areas represent an often-overlooked frontier in the transition to sustainable mobility and land use. Their unique characteristics demand tailored approaches that combine technological innovation with context-sensitive spatial planning. Solutions must bridge the gap between mobility and land development, ensuring that all residents, regardless of location, have equitable access to services, employment and opportunity.

As governments seek to meet climate and development targets, rural communities must be empowered through cross-sector collaboration, community engagement, and flexible policy frameworks. When properly supported, rural areas can serve as models of resilient, decentralized, and sustainable development, harnessing local strengths to meet global challenges.

DISCUSSION

Opportunities

EV technology and charging infrastructure: the emergence of E-Mobility integration

The transition to electric mobility represents one of the most transformative shifts in transport and energy systems in recent decades, with advantages for air quality, climate change, national autonomy. Driven by rapid technological advancement, falling battery costs and growing environmental concerns, EV deployment is expanding at an unprecedented rate. Central to this evolution is the charging infrastructure, which acts not only as a point of vehicle refueling but also as a critical interface between the transportation and energy sectors.

This interface presents a spectrum of opportunities. From accelerating decarbonization and enabling smart energy systems, to supporting inclusive urban development and unlocking new business models, the convergence of EV technology and infrastructure signals a paradigm shift in mobility planning. The integration of e-mobility infrastructure into land use and energy systems offers the potential for a more sustainable, accessible and resilient future.

Technological advancements: foundations for scalable integration

Technological progress in EVs is reshaping the limits of electric mobility. Innovations in battery chemistry are delivering 30-50% more range per charge, enhancing consumer confidence and enabling longer trips (IEA, 2023). Rapid charging technologies, such as 350kW ultra-fast chargers, now reduce recharge times to under 15 minutes, bringing electric refueling closer to the convenience of conventional petrol stations.

One of the most groundbreaking developments is V2G capability, which enables EVs to serve as mobile energy storage units. Through bidirectional power flows, EVs can discharge energy back to the grid, helping to balance supply and demand during peak hours and acting as a decentralized grid asset (IRENA, 2020). This elevates EVs from mere transport devices to active participants in the energy system.

Charging infrastructure is growing to match this technological momentum. With over 290,000 public charging points globally and a 25% annual growth rate, networks are becoming increasingly robust (Bloomberg NEF, 2022). The power capacity of these systems—now reaching 350kW—has expanded the infrastructure's capability to support heavy-duty and high-usage vehicles, including electric buses and commercial fleets.

Sustainable mobility and the energy transition

The shift to electric mobility is not just about eliminating tailpipe emissions. EVs also contribute to clean air, noise reduction, and improved public health outcomes, particularly in congested urban environments. Reduced particulate matter and minimal noise pollution enhance livability and support wider sustainability goals aligned with the Sustainable Development Goals (SDGs) (UNEP, 2021).

Moreover, EVs facilitate the integration of renewable energy sources into transport systems. Vehicles can be charged using solar and wind power, and in turn, act as distributed storage solutions. When combined with smart charging systems, EVs can respond dynamically to grid signals, optimizing load and stabilizing fluctuating renewable output (IRENA, 2020). This flexibility reduces dependence on fossil fuel backup systems and strengthens energy resilience.

The synergy between EV technology and the grid also supports decentralized energy planning. Rural communities, for instance, can leverage EV infrastructure to enhance energy access, provided that investment in rural grid upgrades and targeted charger deployment is prioritized (World Bank, 2021).

Cross-sectoral synergies and innovation

Beyond the energy and transport interface, EV charging infrastructure opens new pathways for smart city development. Modern charging stations are increasingly being equipped with public Wi-Fi, environmental sensors, and lighting controls, turning them into multifunctional urban nodes. This creates opportunities for data collection, service delivery and urban systems integration, with chargers doubling as hubs for digital innovation.

Another critical area is transit connectivity. Locating charging stations at PT terminals supports seamless intermodal mobility and enhances user convenience. Integrated hubs allow users to switch between personal EVs, buses, trains and shared MM options, creating efficient, low-emission urban mobility networks (UN-Habitat, 2015).

The evolving e-mobility ecosystem also enables new business models. Vehicle ownership is gradually giving way to service-based models, such as electric car-sharing, ride-hailing, and subscription services. At the same time, energy utilities are becoming mobility providers, bundling power delivery with EV services and infrastructure development.

Equity, accessibility and location efficiency

To realize the full potential of e-mobility, infrastructure expansion must be equitable and inclusive contributing to systemic efficiency (Buettner et al. 2024). EVs can lower operating costs for users, making them more affordable in the long term. However, underserved and disadvantaged areas often remain excluded from charging infrastructure, creating so-called “charging deserts” (UNECE, 2022). Addressing this requires targeted investments in rural and low-income communities, coupled with policy mechanisms to ensure affordability.

Shared mobility solutions are key enablers in bridging this gap. Electric car-sharing, community-owned EV fleets, and micromobility services provide access to clean transport for those without private vehicles. Integrating these services into affordable housing developments and community hubs enhances mobility equity.

Location efficiency is also essential for infrastructure sustainability. It requires establishing a system of locations as it was discussed in the related chapters. In this vein, aligning EV infrastructure with urban master plans and new developments ensures minimal disruption, enhanced uptake and long-term utility. Grid-optimized locations, chosen based on both power availability and user demand, reduce costs and support broader energy goals.

Conclusion: realizing a seamless and integrated mobility future

The convergence of EV technology and charging infrastructure presents a once-in-a-generation opportunity to reshape mobility and energy systems. With continued innovation, strategic investment, and equitable policy support, EVs can help deliver a seamless, integrated, and sustainable transport future.

To unlock these opportunities, stakeholders must think beyond vehicles and charging points. They must adopt a systems-level approach that includes land use planning, energy policy, digital infrastructure and social inclusion. By doing so, EV infrastructure becomes not just a technological achievement, but a catalyst for resilient, low-carbon and equitable development.

Barriers

Barriers to EV adoption and charging infrastructure

The global momentum toward electric mobility has gained unprecedented pace in recent years. In 2022 alone, over 10 million EVs were sold worldwide – an increase of 55% from the previous year (IEA, 2023). Market shares in early-adopter countries such as Norway have reached as high as 37%. This growth is largely fueled by technological innovation, climate imperatives and supportive government policy. However, despite these advances, widespread adoption of EVs and the deployment of charging infrastructure face persistent and interconnected barriers. These challenges – spanning financial, technical, regulatory and infrastructural dimensions – must be understood and addressed in a systemic and inclusive manner to achieve a just and equitable energy transition.

Financial constraints and cost parity

High upfront costs remain one of the most significant barriers to EV adoption. On average, EVs are priced 20-40% higher than comparable ICE vehicles. Although battery prices have fallen by over 89% since 2010, battery packs still represent a large part of total vehicle costs. These financial burdens are particularly acute in emerging markets and among low- and middle-income households, for whom even subsidized EVs may be inaccessible.

While some regions have introduced tax credits and rebate schemes to ease the financial burden, the lack of consistent and accessible financing mechanisms continues to limit adoption. Innovations in leasing, battery-as-a-service models, and targeted subsidies for low-income users can help bridge the affordability gap. However, widespread parity between EVs and traditional vehicles remains a future milestone rather than a current reality.

Charging infrastructure and accessibility gaps

Concerns over charging availability, commonly referred to as “range anxiety”, are a leading deterrent for potential EV buyers. Although urban areas in developed countries may have dense charging networks, rural regions, low-income communities and informal settlements frequently lack adequate access. Data from the United States shows that low-income areas have 15% fewer public charging stations compared to affluent ones (US DOE, 2022). Additionally, 42% of households lack access to private home charging, especially in multi-unit dwellings or informal housing contexts.

Even fast-charging technology has limitations, with average times of 20–40 minutes to reach 80% capacity. Public stations often operate under fragmented systems, requiring different apps and payment platforms. The absence of interoperable systems frustrates users and reduces confidence in EV infrastructure reliability.

Grid capacity, flexibility and resilience

The increasing demand for EV charging places a substantial burden on electricity transmission and distribution networks. Many existing grids are ill-equipped to handle high-volume charging loads, particularly during peak demand periods. Without timely upgrades, charging infrastructure may contribute to grid instability, blackouts, or forced curtailments (IRENA, 2020).

Solutions such as smart charging, V2G systems, and stationary battery storage offer pathways to grid optimization. These technologies enable better load balancing, renewable integration, and backup supply during outages. However, their adoption is still limited and often hindered by high capital costs and regulatory complexity.

Technological fragmentation and lack of standards

A major technical barrier to EV infrastructure is the absence of global and regional standardization. Currently, users must navigate between multiple connector types, such as CCS, CHAdeMO and proprietary systems like Tesla, as well as a plethora of non-integrated charging networks. More than 50 different charging platforms require unique apps and access protocols, leading to poor user experience and limited compatibility across regions (UNECE, 2022).

Efforts to promote interoperability, protocol integration and harmonized standards are essential. Universal standards can simplify deployment, reduce costs, and increase adoption. They also support seamless cross-border travel – an important factor in regional economic integration, especially in Africa and the European Union.

Policy and regulatory hurdles

Fragmented regulatory frameworks and outdated utility laws pose additional challenges. The installation of public charging stations often requires navigating multiple permitting processes, sometimes involving more than a dozen agencies across different jurisdictions. These delays slow infrastructure deployment and increase costs.

Outdated building codes often lack EV-ready requirements, meaning that new residential and commercial construction misses opportunities for low-cost charger integration. In some regions, utility investment in charging infrastructure is constrained by rules that limit their ability to generate revenue from transport services. Meanwhile, inconsistent incentives across provinces, states or municipalities create market distortions, reducing investor confidence and operational efficiency.

Conclusion: addressing barriers for an inclusive transition

The barriers to EV adoption and infrastructure deployment are multi-faceted and require coordinated, cross-sectoral responses. Expanding financial incentives, updating building and utility regulations, investing in smart grid technologies, and ensuring equitable access to underserved areas are among the most urgent actions. Equally important is the creation of inclusive governance structures that involve communities in infrastructure planning and implementation.

By identifying and overcoming these barriers, governments and stakeholders can ensure that the benefits of the e-mobility transition – reduced emissions, improved air quality and increased energy resilience – are shared broadly and equitably across society.

Assessing limitations of e-mobility, of location efficiency, of participation

The intersection of mobility and energy

The global momentum for e-mobility is a cornerstone of the sustainable transport and energy transition. As EVs become more common and the supporting infrastructure grows, integration into wider transport and energy systems remains uneven. The technological advances of EVs have enabled new paradigms for power generation, distribution, and consumption. Yet, these benefits are not universally distributed, and many communities remain disconnected from e-mobility's promise. Location efficiency and stakeholder participation are equally critical elements of this transformation but continue to encounter practical, economic and structural constraints. This chapter examines the key limitations of e-mobility systems, location efficiency strategies, and stakeholder participation, offering a more grounded perspective on the complexity of the mobility transition.

Limitations of E-Mobility: technology, infrastructure and equity

Despite growing EV adoption globally, significant barriers persist in delivering widespread, equitable benefits. A primary challenge is the range and charging constraint. Although EV technology continues to improve, range anxiety remains a major deterrent, particularly in regions where long-distance travel is common or where charging stations are sparse. Fast-charging networks have expanded but remain insufficient for high-volume or remote travel needs (IEA, 2023).

Cost remains a critical barrier. High upfront purchase prices, particularly in low- and middle-income economies, restrict EV access to wealthier individuals. Even where government subsidies exist, battery replacement costs and maintenance uncertainties may put in question long-term affordability (Bloomberg NEF, 2022). Additionally, grid infrastructure is under strain. As EVs scale, electricity networks must accommodate increased peak demand. Many grids, especially in emerging economies, lack the resilience or capacity to support widespread, high-power EV charging. Without smart charging and grid upgrades, widespread e-mobility adoption could exacerbate existing energy system vulnerabilities (IRENA, 2020).

The environmental impact of e-mobility also warrants scrutiny. While EVs emit no tailpipe pollutants, their batteries depend on minerals like lithium and cobalt, which are often extracted under socially and environmentally exploitative conditions. Recycling systems for EV batteries remain underdeveloped, risking future environmental degradation and material scarcity (World Bank, 2020).

Constraints in location efficiency: the space–energy–behaviour nexus

Location efficiency—the strategic placement of services, infrastructure, and housing to reduce mobility needs—offers a key pathway to sustainable urban systems. However, realizing these efficiencies is increasingly difficult. In dense urban environments, limited space leads to land use competition. Charging infrastructure must often compete with essential public services, green spaces, and affordable housing. Without deliberate integration into urban design, EV infrastructure may exacerbate space scarcity and gentrification (UN-Habitat, 2020).

Optimal placement of charging infrastructure also depends on user behaviour and demand patterns. Unfortunately, current charging behaviours do not always align with energy efficiency or grid optimization goals. Many users prefer to charge at home or at convenient locations—habits that may conflict with grid balancing needs or sustainable land use planning.

Power grid integration adds another layer of complexity. Even when land is available, extending or upgrading electricity networks to reach desirable sites for charging infrastructure may be cost-prohibitive or technically challenging. In many jurisdictions, infrastructure deployment occurs without sufficient coordination between energy, land use and transport authorities, reducing overall system efficiency (UNECE, 2022).

Accessibility and participation limitations: who is being left out?

Despite the promise of e-mobility, the distribution of infrastructure remains highly unequal. In many countries, charging stations are heavily concentrated in affluent urban centres, while low-income communities and rural regions are left behind. Companies and investors typically prioritize high-traffic or high-income areas, creating "charging deserts" in regions with limited profit potential. For rural residents, sparse infrastructure and long travel distances exacerbate range anxiety and make EV ownership impractical (World Bank, 2021).

Beyond geography, user diversity poses further challenges. Charging systems often overlook the needs of persons with disabilities, older adults, or the digitally excluded. Payment systems may not support

cashless-unbanked populations, and EV education or interface instructions are frequently unavailable in multiple languages. Such limitations hinder widespread adoption and deepen existing transport inequalities (UNEP, 2021).

Participation in mobility planning also faces structural obstacles. Many planning processes exclude grassroots input, focusing instead on technical or commercial stakeholders. Where engagement does occur, it is often superficial, failing to meaningfully influence infrastructure design or deployment. Knowledge gaps about e-mobility among the general public persist, fueled by misinformation, a lack of targeted outreach, and minimal representation in decision-making forums (UNECE, 2022).

Towards integrated solutions: innovation, policy, and collaboration

While the challenges are substantial, solutions are emerging. Technological innovation continues to address many limitations. Advancements in battery energy density are gradually alleviating range concerns, while V2G technologies are creating new pathways for energy sharing and grid stability (IEA, 2023).

Policy reform is also critical. Equity-focused incentives – such as subsidies for low-income EV buyers, mandatory rural charging deployment, and zoning reforms for integrated infrastructure – can address current disparities. Updated building codes requiring EV-ready infrastructure, as well as public financing for strategic infrastructure, are essential to support inclusive access (UN-Habitat, 2022).

Most importantly, cross-sectoral collaboration must become the norm. Planning and deployment of e-mobility infrastructure should involve energy regulators, urban planners, community leaders and private sector actors. Public-private partnerships can accelerate development timelines, while academic research can provide the evidence base to inform adaptive, just and resilient strategies.

Conclusion

E-mobility, location efficiency, and public participation each hold transformative potential for creating sustainable transport and energy systems. Yet, their limitations – ranging from cost and infrastructure to access and governance – must be addressed with urgency and precision. Achieving the objectives of inclusive, low-emission mobility will require not only technology and investment, but deliberate structural reforms in how we plan, prioritize and engage in shaping the transport systems of tomorrow.

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ANNEX – Electric mobility in Sweden: consumer behaviour and regional spread of battery electric vehicles in 2016-2022

This case study is a summary of a section from a larger study on EVs in Sweden. The full report is available at: <https://www.trafa.se/vagtrafik/elektrifierade-fordon-i-sverige--en-analys-av-laddbara-fordon-over-tid-och-geografi-15113/>.

Introduction

The global transition to EVs has gained momentum in recent years, and Sweden has emerged as a significant case study due to its ambitious climate goals and policies supporting EV adoption. Since 2020, the sales of BEVs in Sweden have grown rapidly, with batteries becoming the predominant energy source for newly registered vehicles by 2022. Despite this impressive growth we know little about the distribution of EV ownership among different household groups. In this study, utilizing data from Sweden to shed light on the burgeoning EV market, we investigate the demographics of predominantly early EV adopters. The aim of this study is to provide a better understanding of which Swedish households have been early adopters of EVs, and examine how these differ from those who choose ICE cars. We will also investigate potential differences between private purchases and private leasing, as well as how the geographic distribution of BEVs across the country has evolved over time.

The literature on the interest in buying EVs' goes back to about 2010, coinciding with the broader market introduction of EVs. Most of the previous studies, especially the earlier ones at the time when the market for EVs was still small, have focused on attitudes and interests, of individuals or households, towards the possibility of buying an EV in the near future. The literature has since then evolved into a broader palette of research questions concerning EVs. The main method is still to this day studies of stated preference. Some studies have also combined various data sources—such as interviews and surveys – with GPS data from vehicles and charging infrastructure to gain a better understanding of how the vehicles are used. Revealed preference studies are rare and registry data are very seldom used.

In later years, the research has evolved into more data driven studies, although the literature continues to be dominated by surveys and interview studies on intended future purchases. We have found only a few studies that are based on more extensive data, from Sweden and Norway.

In our literature review we have also found a few studies with a particular focus on identifying 'early adopters', i.e. individuals who are interested in new technologies and willing to opt for an electric vehicle early on. Ploetz et al. used revealed preference and found some interesting discrepancies between the common assumption that EVs attract especially younger, environmentally interested and technically savvy people. Instead, Ploetz found that the group of early EV purchasers in Germany mainly consisted of men aged 40-50 who live in a house with multiple family members, enjoying a high financial standard of living.

In this study, we aim to identify the primary consumer segments contributing to BEV adoption in Sweden and examine how the geographic distribution of these vehicles has evolved over time. By analyzing the behavior of both private individuals and companies, we explore the purchasing and leasing patterns of BEVs, offering insights into the evolution of electric mobility in the context of socio-economic factors. Specifically, in our regression analysis we extend earlier research on the adoption of

new BEVs in 2016 and 2020, incorporating a broader dataset from 2020 to 2022, also distinguishing between new and used BEVs. This expanded scope allows us to investigate whether distinct household characteristics influence the choice of new versus used BEVs and the implications for market saturation and policy intervention. The study aims to further develop previous research findings on the driving mechanisms behind households' consumption of new cars and BEVs.

Data and method

In this study we use two interconnected registry databases from Statistics Sweden. The first database covers all individuals registered in Sweden 2016, 2020 and 2022, featuring diverse socio-economic variables. The second database details all privately owned vehicles in Sweden, with provisions for linking vehicles to their owners in the individual-level database. The data also includes cars that are leased or are company cars but used by the household. The definition of a leased vehicle is that it needs to be leased for at least one year and be registered to an individual user.

By matching vehicle data (4.6 million vehicles) with data on Swedish households (4 million), we were able to link car ownership with several socio-economic factors. These observations were in turn linked to small geographical areas (DeSO) for 2016, 2020 and 2022 to match additional information to each household. We discard households owning more than five cars ($<0.03\%$) from the analysis. We also discard households having more than 12 household members ($<0.01\%$).

In 2020, BEVs accounted for 9% of all newly registered passenger cars in Sweden. By 2022, this share had increased to 32%. Initially, BEVs were primarily adopted by companies, but from 2020 onwards, private individuals also began purchasing and leasing BEVs. In 2020, approximately 4,300 BEVs were purchased by private individuals, and just over 5,000 were acquired through private leasing. Together, these accounted for 33% of all newly registered BEVs that year. By 2022, the number of privately purchased BEVs had risen to nearly 14,000, and the number acquired through private leasing had reached almost 35,000, amounting to 51% of all newly registered BEVs.

However, the transition of vehicles already in use has progressed towards electricity much more slowly. Sweden has nearly five million passenger cars in operation. Of these, only 1.1% were BEVs in 2020, a figure that had increased to 4% by 2022.

In the analysis we estimate five regression models for the year 2020 and 2022. The analysis was done in a sequence of steps. In the first step, Model 1 is estimated to sort out relevant factors explaining why households register a new vehicle, irrespective of the kind of fuel type. The following four models (Model 2-5) use the stock of households that have registered at least one vehicle in 2020 and 2022 (taken from Model 1) to analyze factors of importance to households choosing EVs and the choice to buy or lease these cars.

Step two involved a detailed analysis of each variable's association with the dependent variable, both individually and in conjunction with other explanatory factors. This was done using scatter plots to investigate which variables had a positive or negative relationship with the dependent variable.

Regression analysis

The results reveal a marked increase in the adoption of BEVs in Sweden between 2016 and 2022. While ownership has expanded across the population, the transition remains unevenly distributed across socioeconomic and demographic groups. Income is the most significant predictor of BEV adoption. The probability of purchasing a BEV increases substantially with income, and this effect has grown

stronger over time. In 2022, individuals in the highest income deciles were far more likely to purchase a BEV than those in the lower deciles. Leasing is also positively correlated with income, but the effect is less pronounced than for purchases. This suggests that while leasing lowers the financial threshold for BEV adoption, it does not eliminate structural inequalities.

Housing type plays a central role. Individuals living in detached houses are significantly more likely to both purchase and lease BEVs. The effect is stronger for purchases, indicating that homeownership and the ability to install private charging infrastructure remain key enablers. Residents in multi-dwelling units are underrepresented, particularly in the ownership segment. Household composition also matters. Couples, especially those with children, have lower odds of both purchasing and leasing BEVs compared to single-person households. The likelihood of leasing among single adults is particularly low, pointing to a gap in accessibility for smaller or less resource-rich households.

Age exhibits a curvilinear effect. Middle-aged individuals (36-55) are the most likely to purchase EVs, while both younger (<35) and older (>65) age groups have significantly lower adoption rates. Leasing appears somewhat more accessible to younger age groups, though their overall representation remains modest. In geographic terms, urban and suburban municipalities show higher levels of leasing, while EV purchases are more common in affluent suburban and semi-rural areas. Rural municipalities lag behind significantly in both categories. While EV ownership is spreading, these results confirm that structural differences in adoption remain strong. Leasing has allowed new user groups to access EVs, but the broader pattern of socioeconomic stratification is largely intact.

Differences between new registrations and vehicles in use

While both new registrations and vehicles in use provide valuable insights into the electrification of the Swedish car fleet, the two perspectives reveal different patterns of ownership and transition dynamics.

First, socioeconomic disparities are more pronounced in new registrations than in the stock of vehicles in use. Income, for instance, has a stronger effect on the likelihood of owning a newly registered EV than on owning one that is already in traffic. This indicates that structural inequalities are more visible at the front line of the transition, where new technologies and market trends take hold. Over time, as these vehicles become part of the broader vehicle stock, the differences between groups become less distinct.

Second, age-related patterns are more prominent in new registrations. Younger (<35) and older (>65) individuals are significantly underrepresented among new EV owners. However, these effects are weaker in the vehicle fleet currently in use. This is likely due to the long lifespan of vehicles, which allows cars initially registered by one demographic group to later be owned by another, smoothing out the demographic distinctions.

Geographic and ownership transitions of newly registered BEVs

To complement the regression analysis of socio-demographic predictors of EV adoption, we also conducted a longitudinal flow analysis of BEVs registered in 2020. The aim was to examine how these vehicles have moved across geographic and ownership categories over a four-year period. The analysis includes both privately and corporately owned vehicles.

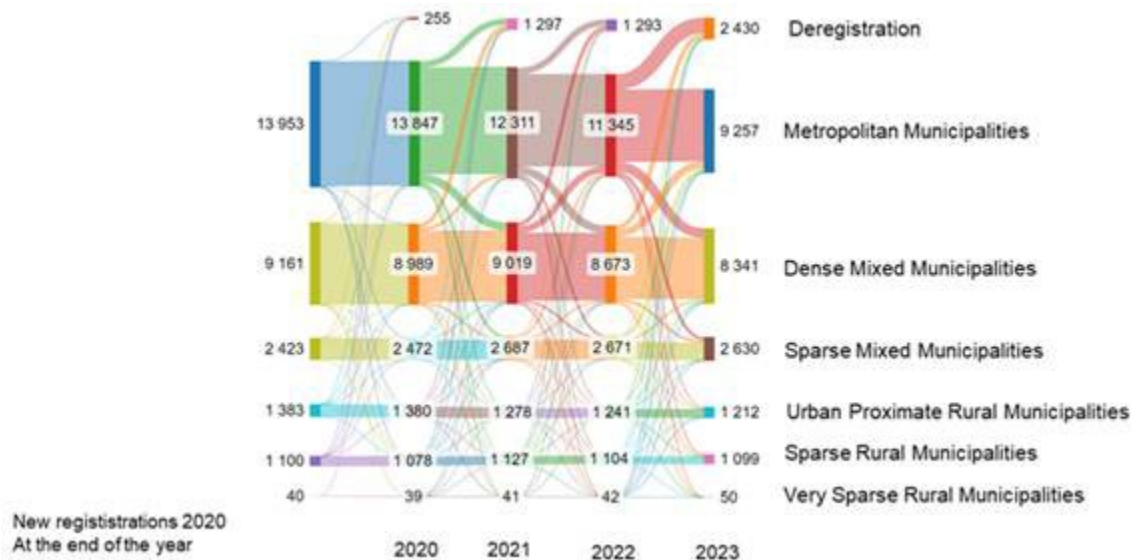
Using the six municipal categories defined by the Swedish Agency for Economic and Regional Growth, we tracked the geographic location of each BEV at the time of registration in 2020 and in subsequent years until the end of 2023. Most vehicles were initially registered in metropolitan municipalities

(Stockholm, Gothenburg, Malmö) and in dense mixed municipalities located in southern and central Sweden – 13,953 and 9,161 vehicles, respectively.

The majority of these vehicles remained within the same municipal category over time. By the end of 2023, 9,257 vehicles were still registered in metropolitan municipalities and 8,341 in dense mixed municipalities. While there was a notable flow of vehicles between these two urban categories, very few BEVs had moved into more rural or sparsely populated areas.

Figure A.1

Newly registered BEVs in 2020 by the regional affiliation of the owner in Sweden



Source: According to the Swedish Agency for Economic and Regional Growth's municipal classification), at the time of registration and at the end of 2020, 2021, 2022, and 2023, respectively.

Instead of gradually diffusing into other municipal types, a significant portion of the BEVs were deregistered, particularly in 2023. Of all BEVs registered in 2020, 19% had been removed from the registry by 2023, primarily through export. The number of deregistered vehicles in 2023 alone was equivalent to the entire BEV stock in rural municipalities, underscoring the magnitude of this outflow.

The limited geographic spread of BEVs to rural areas can be partially attributed to differences in the age profile of the entire vehicle stock. The average age of passenger cars in sparsely populated municipalities was 14–17 years in 2023, making it unlikely that recently registered BEVs—typically four years old or younger—would appear in these areas. As BEVs age and change owners, broader spatial diffusion may occur in the future.

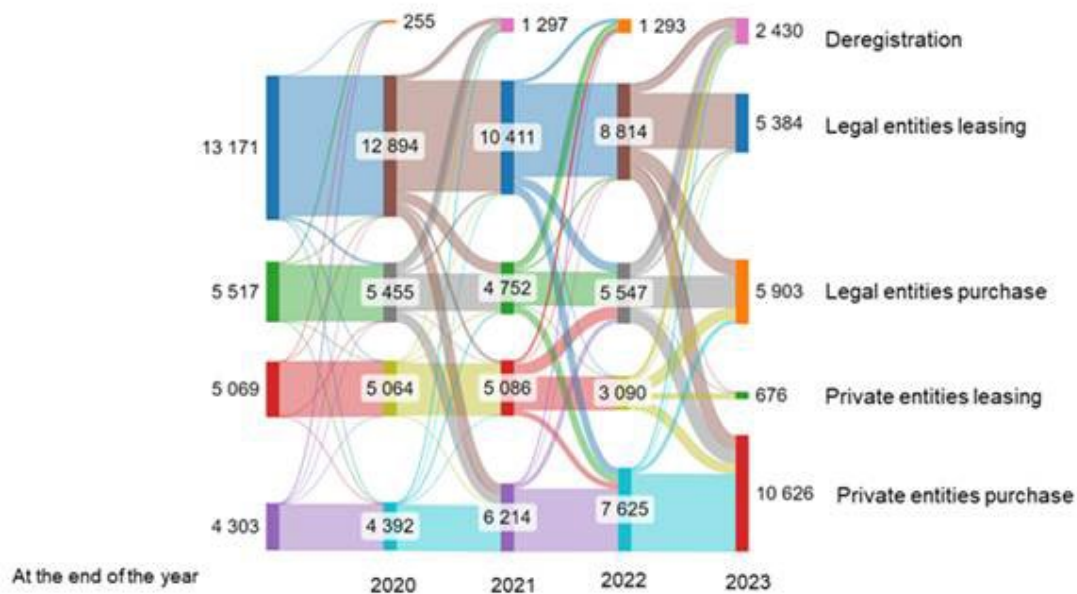
Shifting ownership patterns: leasing and purchase dynamics

A second flow analysis focused on changes in ownership type over time, distinguishing between private individuals and corporations, and between leasing and purchase. Among BEVs registered in 2020, the number owned outright by private individuals more than doubled by 2023 – from 4,303 to 10,626. This increase was made possible by the transition of vehicles from corporate leasing into the private second-hand market, particularly after the typical three-year leasing period had ended.

Meanwhile, the number of BEVs leased by private individuals declined sharply – from over 5,000 in 2020 to just 676 in 2023. Similarly, the number of vehicles leased by legal entities (corporations) was halved over the same period, dropping to 5,384. While the number of vehicles purchased by legal entities increased slightly in 2023 (from 5,517 to 5,903), the general trend indicates a substantial ownership shift from leasing to private purchase.

Figure A.2

Newly registered electric passenger cars in 2020 by ownership type (private/legal entity, purchase/leasing/deregistration), at the time of registration and at the end of 2020, 2021, 2022, and 2023, respectively



This pattern highlights the key role of the corporate sector in introducing BEVs to the market. Leasing through companies serves as an initial entry point, after which vehicles become available to private individuals through the second-hand market. The transition from corporate leasing to private ownership underscores the importance of company cars in facilitating broader diffusion among households.

The geographic flow analysis reveals a highly concentrated pattern of BEV retention in urban areas, with minimal diffusion into rural or peripheral municipalities. At the same time, the ownership analysis points to a dynamic reshaping of the BEV stock: as leases expire, vehicles shift from corporate to private hands. However, this change in ownership does not necessarily coincide with a change in geographic location.

These results suggest that while the ownership base of BEVs is broadening, the spatial distribution remains narrow. The corporate leasing market plays a vital role in initiating the adoption cycle, but current patterns indicate that the benefits of electrification remain largely urban. Over time, further diffusion may occur as vehicles age further, but policy intervention is likely needed in order to accelerate and support this transition.

Discussion

The transition to electric mobility in Sweden is progressing rapidly, yet the benefits are not equally shared across all household types. The results indicate that BEVs are primarily adopted by high-income households, individuals living in detached houses, and middle-aged couples. Leasing has somewhat widened access, but its reach remains socially and geographically constrained. From a policy perspective, this raises important challenges. If the goal is to accelerate the decarbonization of the vehicle fleet while promoting social inclusion, targeted interventions are needed. Several groups are currently underrepresented among EV adopters: low-income households, single adults, apartment dwellers, and rural residents. These groups face financial, infrastructural, and informational barriers that are not adequately addressed by current instruments.

Leasing plays an important intermediary role. It reduces the upfront costs of EV access and may appeal to individuals who are unwilling or unable to commit to a full purchase. However, leasing alone is not a panacea. Even leased EVs are disproportionately adopted by wealthier households with stable housing and relatively high income. Thus, policies that aim to expand leasing must be complemented by broader measures—such as means-tested subsidies, investments in public and shared charging infrastructure, and tailored outreach to underrepresented groups.

Another policy implication is the importance of housing-based strategies. Since individuals living in detached houses dominate BEV adoption, expanding access among apartment dwellers is critical. This may involve regulatory requirements for charging infrastructure in new and existing buildings, or support for neighbourhood-based charging hubs. In summary, leasing has made EVs more accessible, but socioeconomic and spatial inequalities persist. A more inclusive EV transition will require deliberate policy action to reach beyond the early adopter segments and ensure that all households – regardless of income, location, or housing type – can participate in and benefit from the electrification of transport.

Finally, from a policy standpoint, it is crucial to distinguish between adoption patterns in new registrations and the actual composition of vehicles on the road. Policies aimed solely at stimulating new EV purchases may not immediately impact the broader vehicle fleet or address long-term disparities. To ensure an equitable and comprehensive transition, strategies must consider both entry into the system (new registrations) and the legacy fleet (vehicles in use), acknowledging that system-wide change is gradual and affected by second-hand markets and vehicle turnover rates.

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Location efficiency and accessibility of E-Mobility Mobility needs and infrastructure planning

The transition to electric mobility requires more than the deployment of electric vehicles, and calls for a coordinated planning across transport systems, charging infrastructure, energy networks, and land-use. Recognizing these interlinkages, in 2024 the United Nations Economic Commission for Europe established the Informal Task Force on Electric Mobility as a multidisciplinary initiative to support a balanced and resilient integration of electric mobility within broader energy and mobility systems.

This publication presents the research outcomes of Cluster E, which explored the concept of location efficiency and accessibility and its implications for energy infrastructure planning and mobility demand. The report examines the links between e-mobility, land use, governance, behaviour, and investment, highlighting how integrated planning can reduce mobility needs while improving accessibility and system efficiency – connecting buildings, industry, transport systems, and energy infrastructure.

Drawing on expert consultations across expert bodies, the study provides insight into infrastructure requirements, behavioural factors, governance approaches, and policies shaping the future of sustainable mobility. Addressing opportunities and barriers, it contributes to ongoing efforts to develop more efficient, resilient, and inclusive mobility systems across the UNECE region.

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